

Comparative ADHD-Control Signal Processing Analysis Report

1. Methodology and Technical Workflow

The objective of this analysis was to implement a robust signal processing pipeline for a 19-channel EEG dataset using the 10-20 international standard. The workflow was executed as follows:

- **Group Segmentation:** The dataset was split into two primary cohorts: ADHD_Positive (N=61) and ADHD_Negative/Control (N=60).
- **Frequency Filtering:** A dual-filter approach was applied to all channels. A 50 Hz IIR Notch Filter was used to eliminate powerline interference, followed by a 4th-order Butterworth Bandpass Filter (1–40 Hz) to isolate physiological brainwaves.
- **Artifact Rejection via FastICA:** Independent Component Analysis (ICA) was performed on 30-second epochs. A correlation heuristic identified the component most associated with the Fp1 channel to specifically target and zero out ocular artifacts (blinks).
- **Advanced Spectral Visualization:** For each channel, a 2×2 or 4×2 diagnostic grid was generated, comparing raw vs. processed states across Time, Frequency (FFT), Power Spectral Density (PSD), and Spectrogram domains.

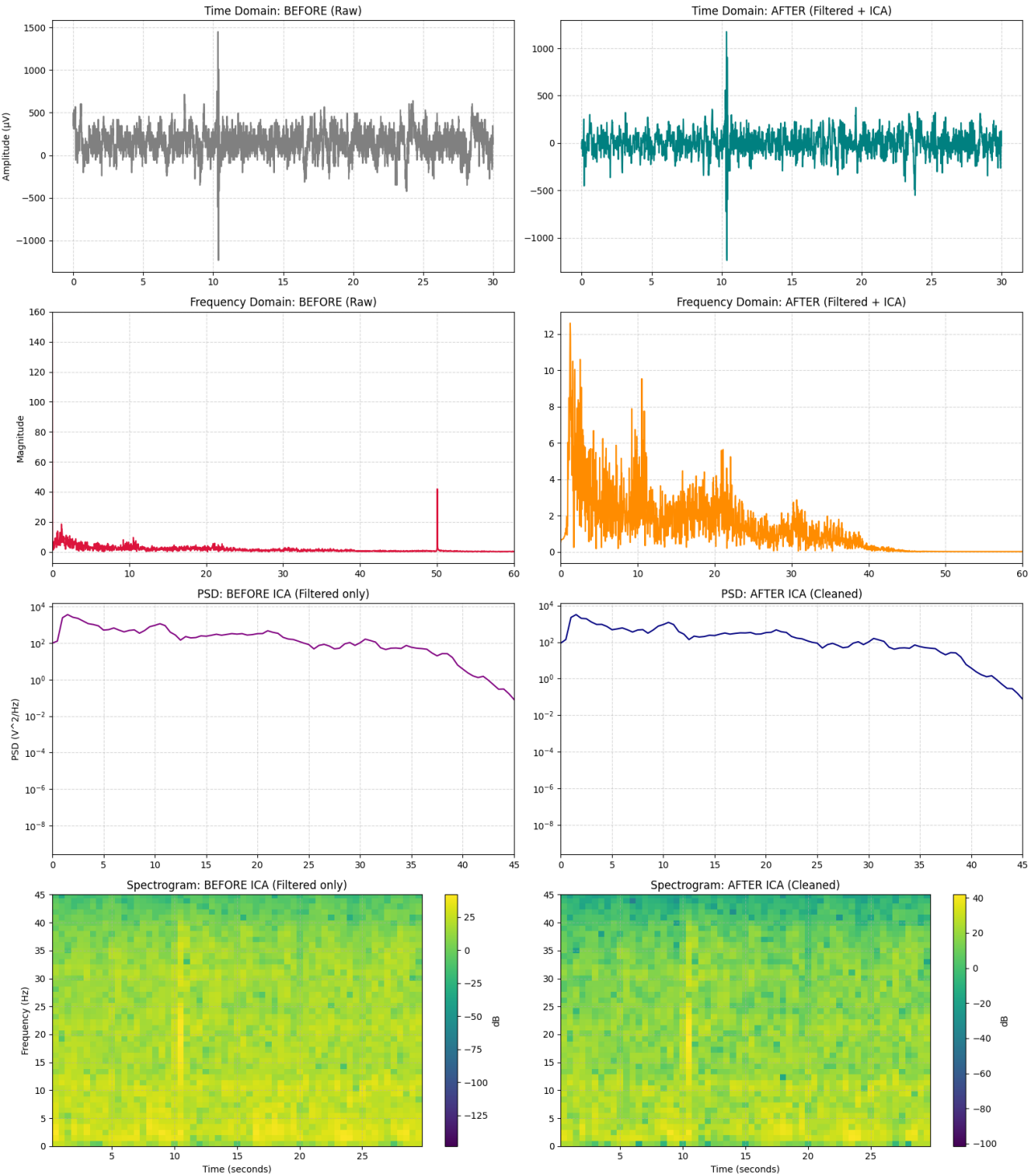
Channel-by-Channel Comparative Analysis

The following section presents each of the 19 EEG channels side-by-side: ADHD Positive group first, followed by ADHD Negative (Control) group, enabling direct morphological comparison.

Channel: T8

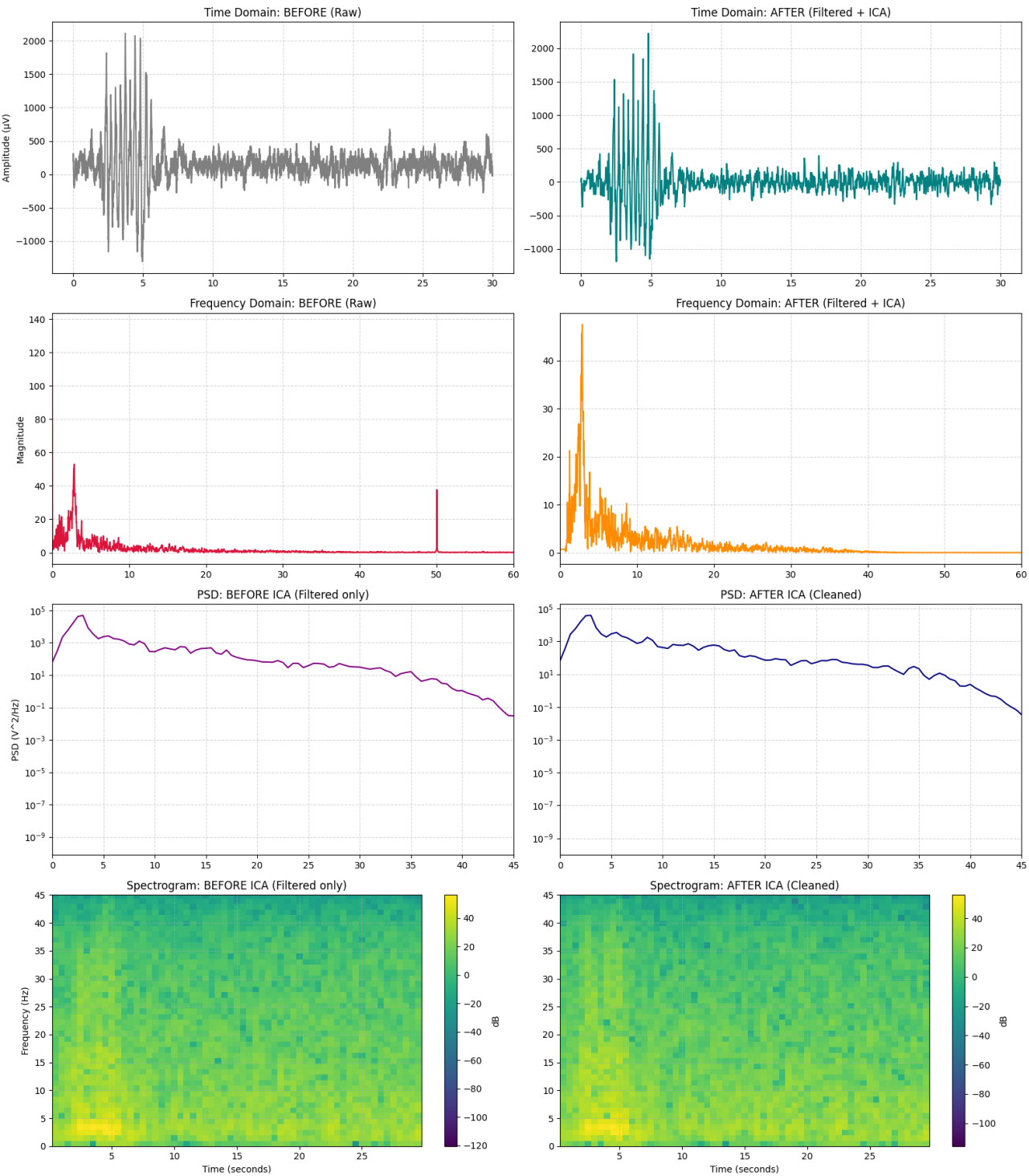
ADHD Positive — Channel T8

Channel: T8 | Group: ADHD_Positive



ADHD Negative (Control) — Channel T8

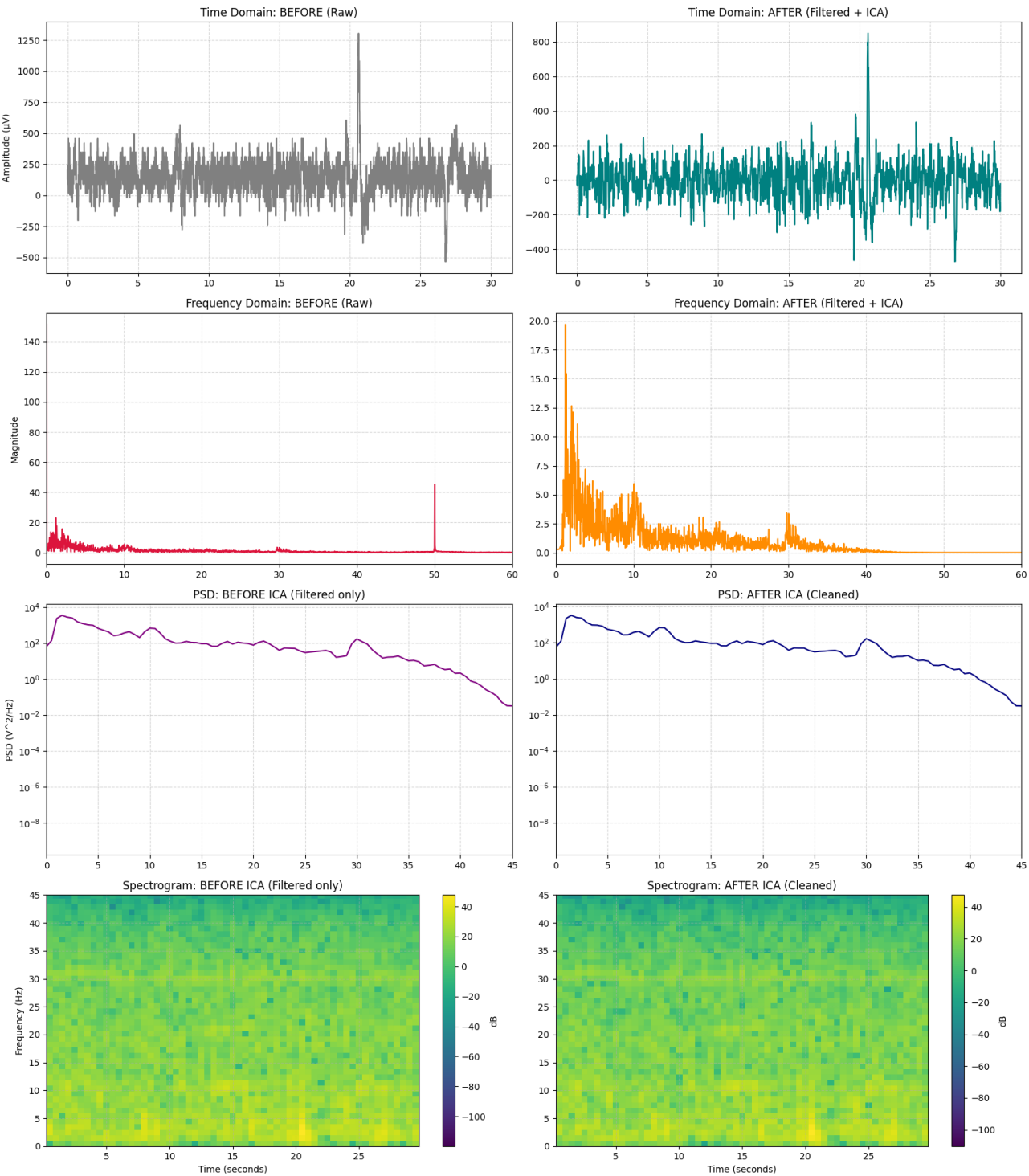
Channel: T8 | Group: ADHD_Negative



Channel: T7

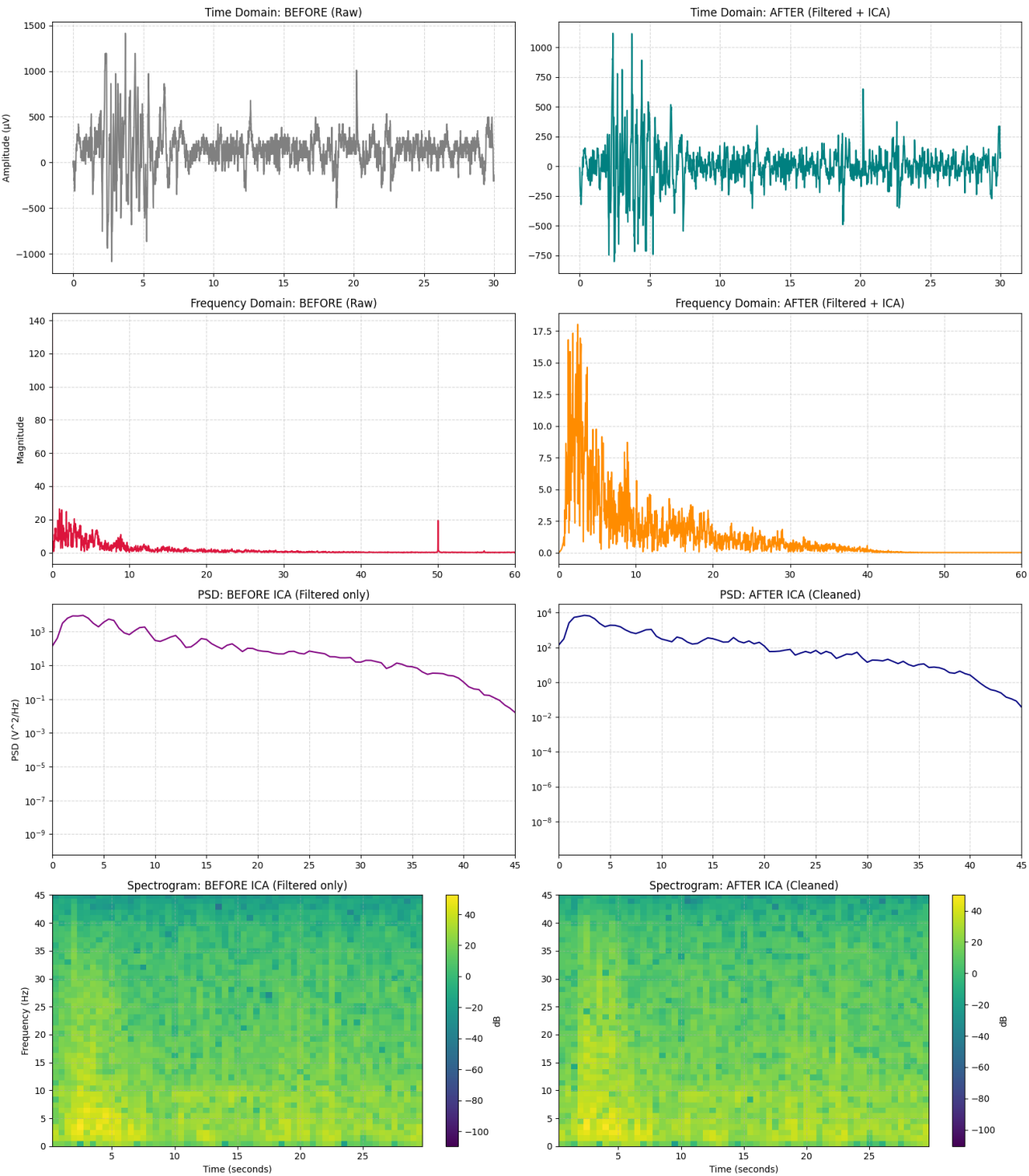
ADHD Positive — Channel T7

Channel: T7 | Group: ADHD_Positive



ADHD Negative (Control) — Channel T7

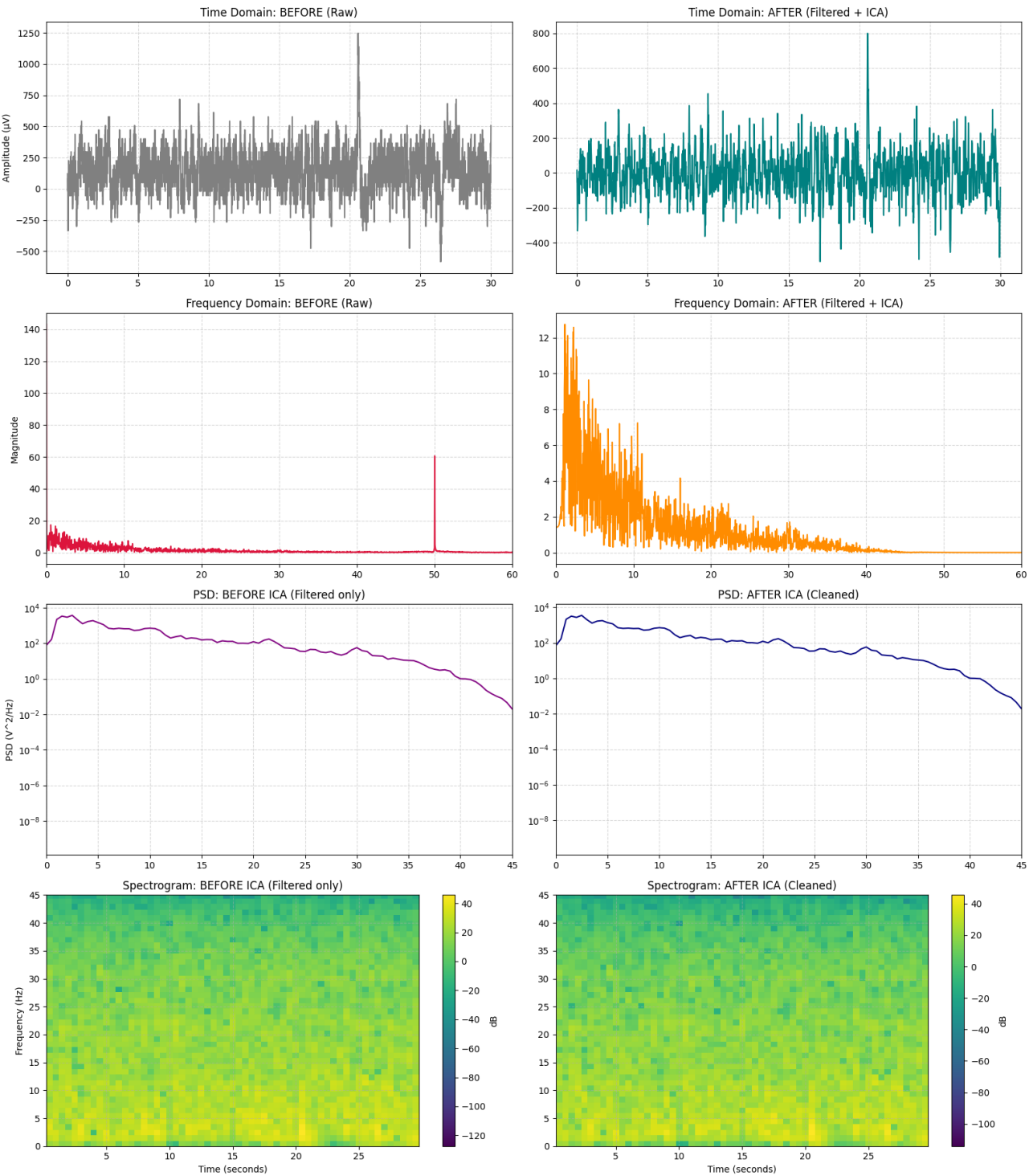
Channel: T7 | Group: ADHD_Negative



Channel: Pz

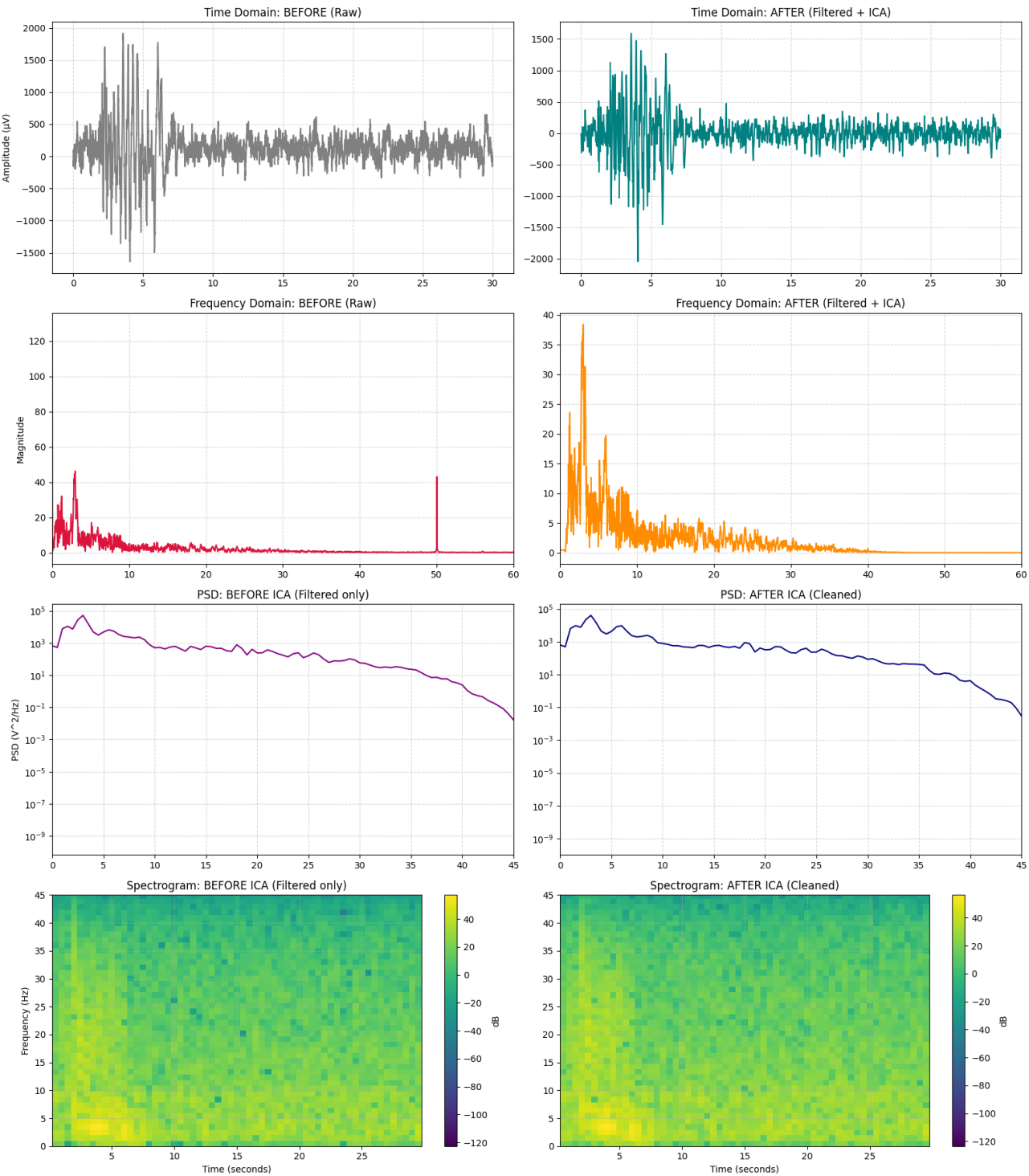
ADHD Positive — Channel Pz

Channel: Pz | Group: ADHD_Positive



ADHD Negative (Control) — Channel Pz

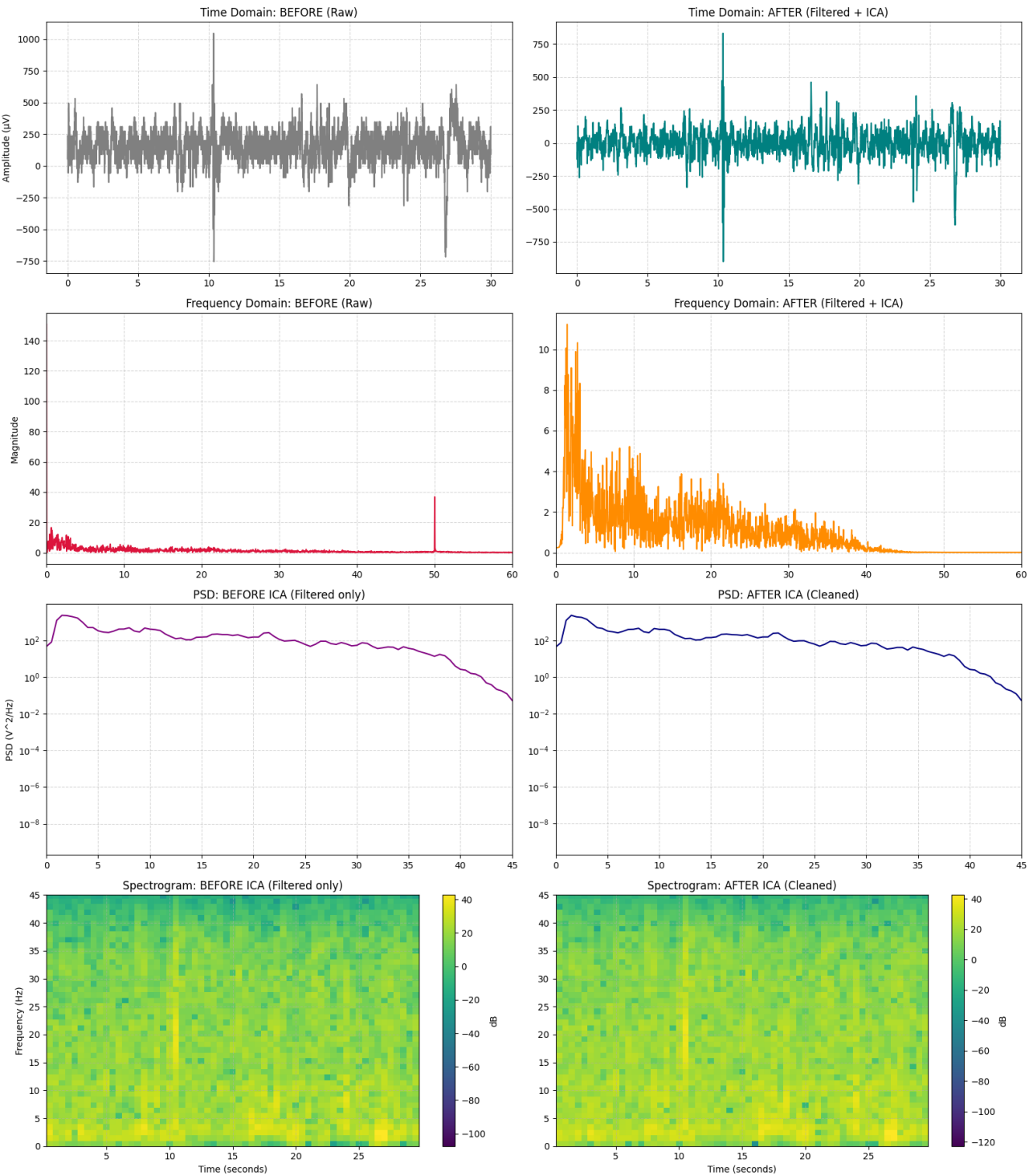
Channel: Pz | Group: ADHD_Negative



Channel: P8

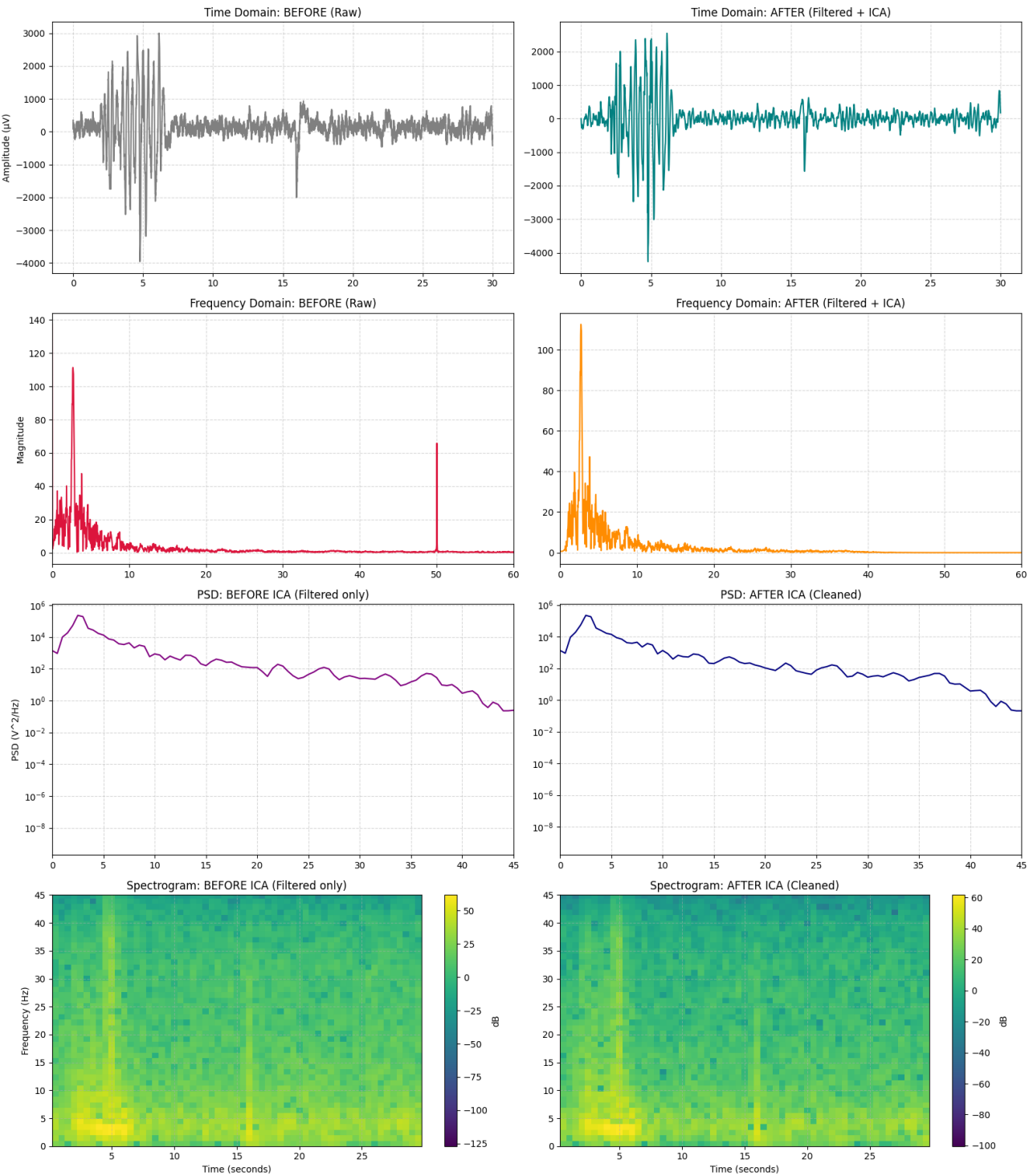
ADHD Positive — Channel P8

Channel: P8 | Group: ADHD_Positive



ADHD Negative (Control) — Channel P8

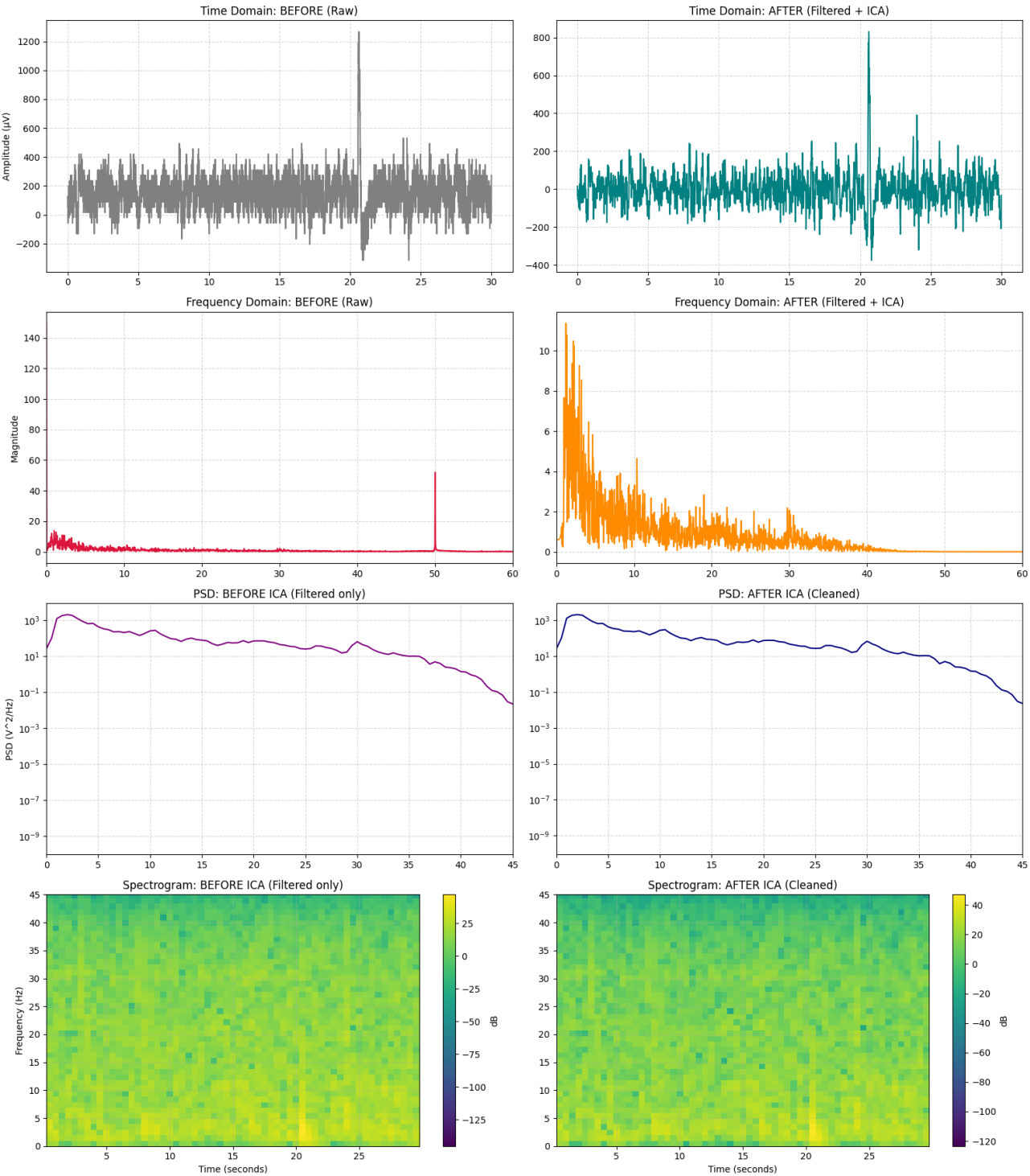
Channel: P8 | Group: ADHD_Negative



Channel: P7

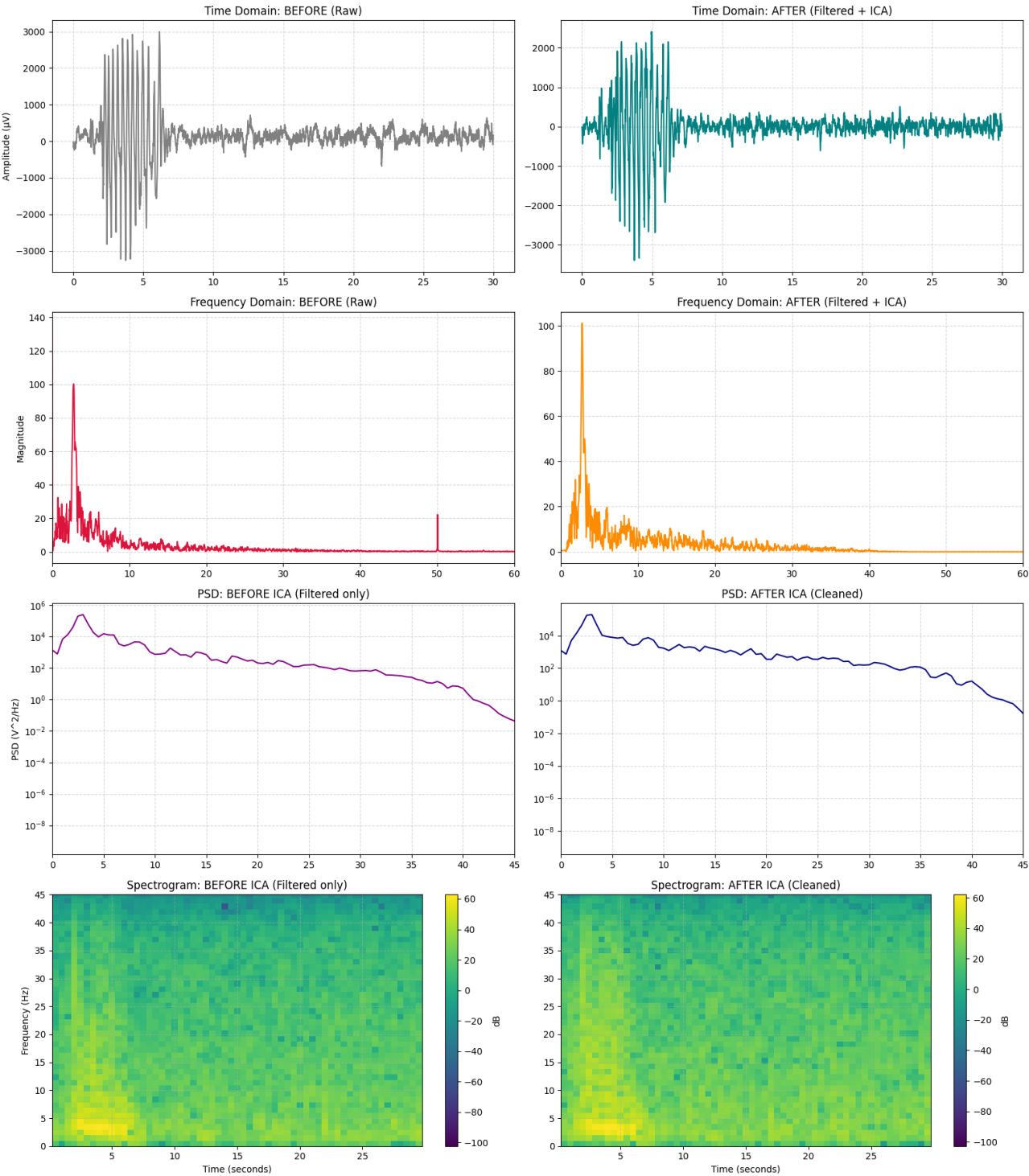
ADHD Positive — Channel P7

Channel: P7 | Group: ADHD_Positive



ADHD Negative (Control) — Channel P7

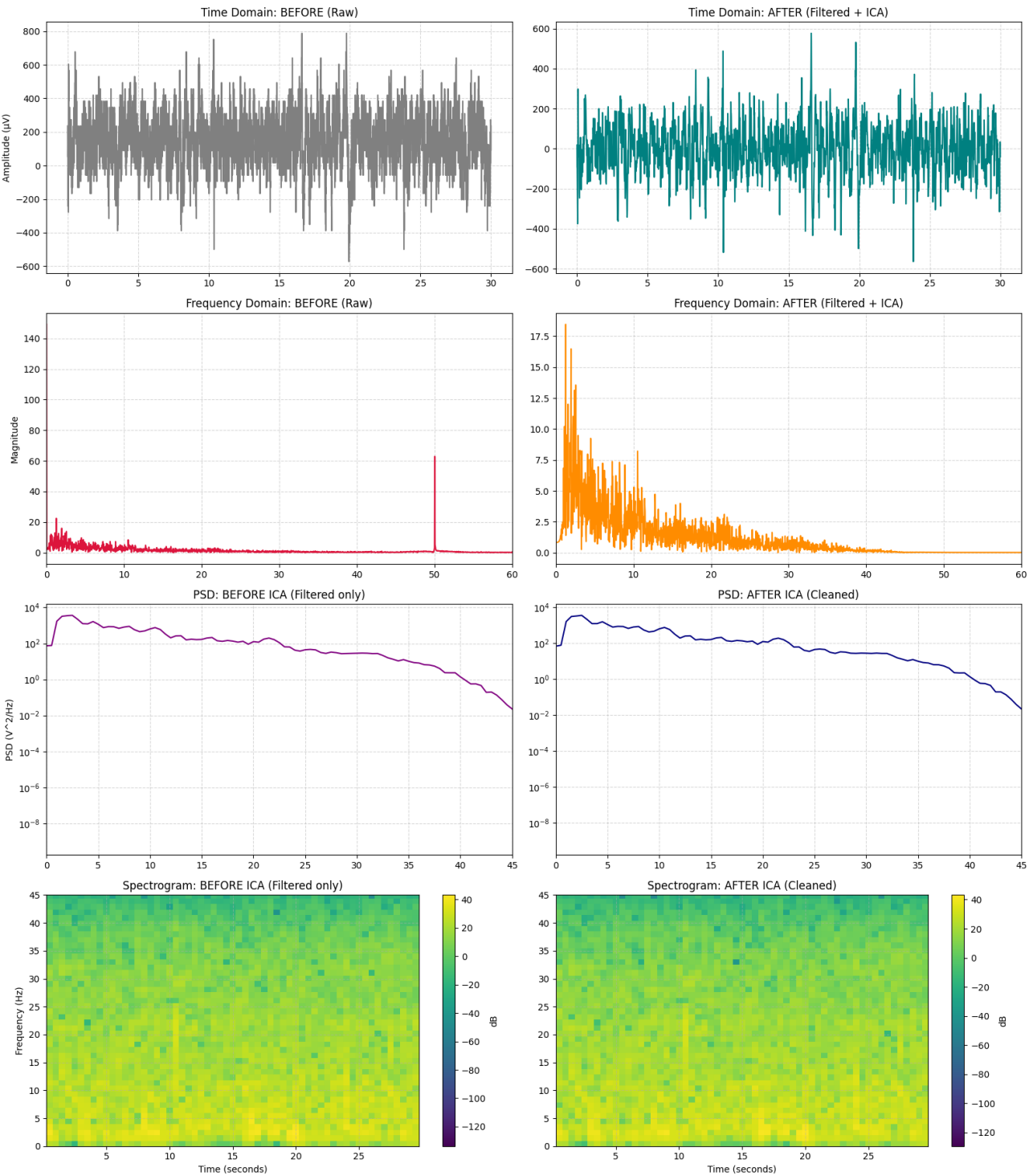
Channel: P7 | Group: ADHD_Negative



Channel: P4

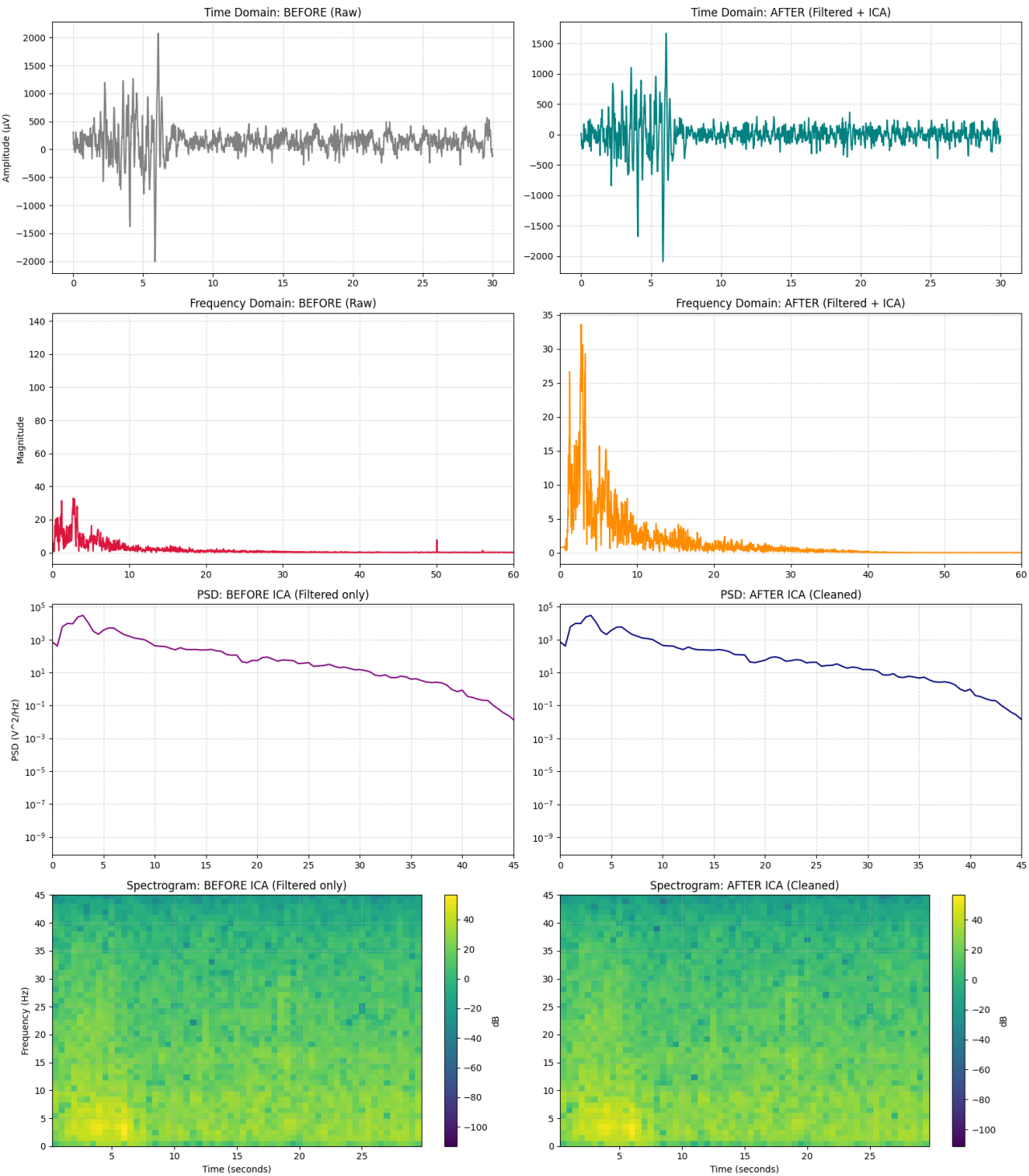
ADHD Positive — Channel P4

Channel: P4 | Group: ADHD_Positive



ADHD Negative (Control) — Channel P4

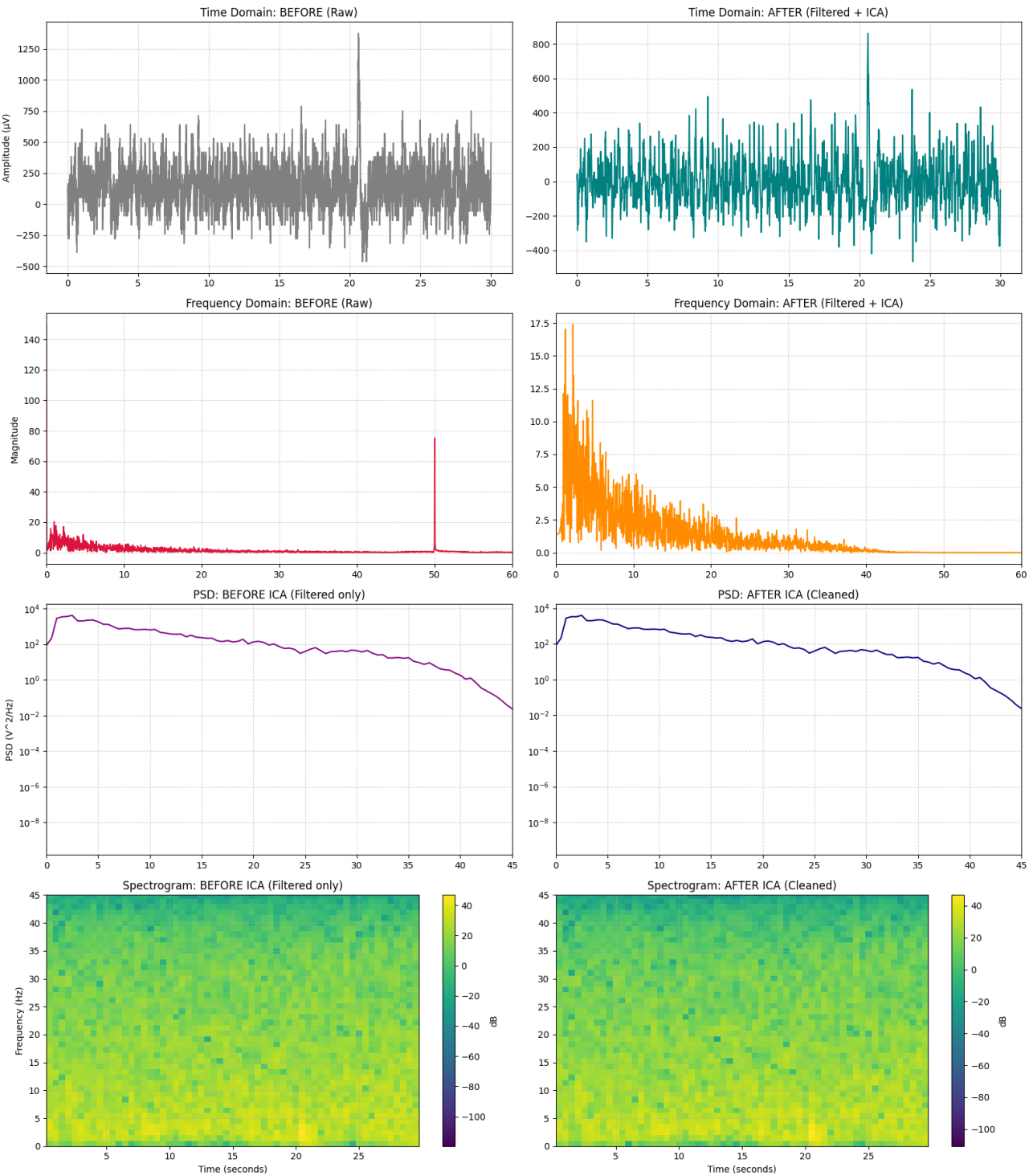
Channel: P4 | Group: ADHD_Negative



Channel: P3

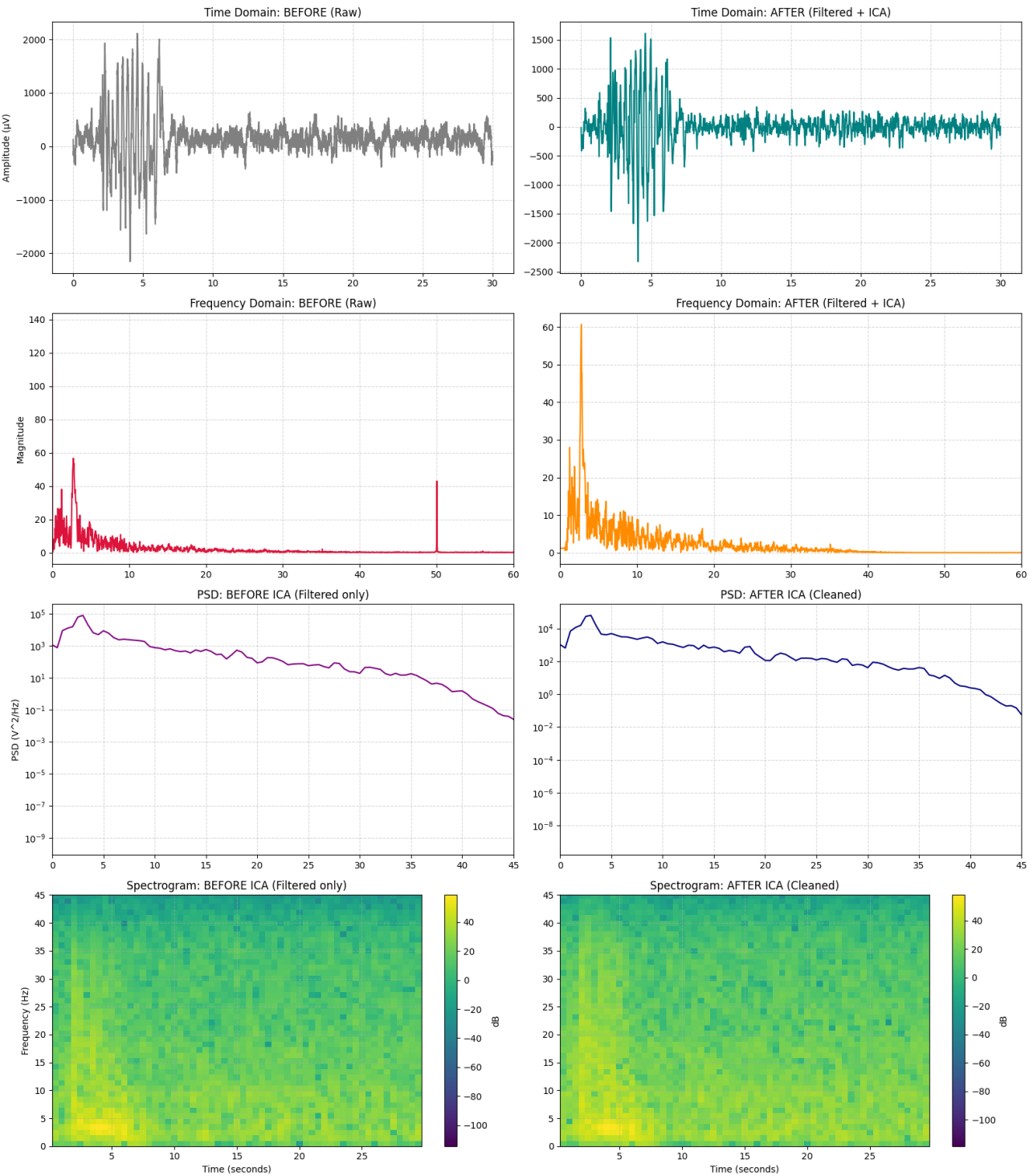
ADHD Positive — Channel P3

Channel: P3 | Group: ADHD_Positive



ADHD Negative (Control) — Channel P3

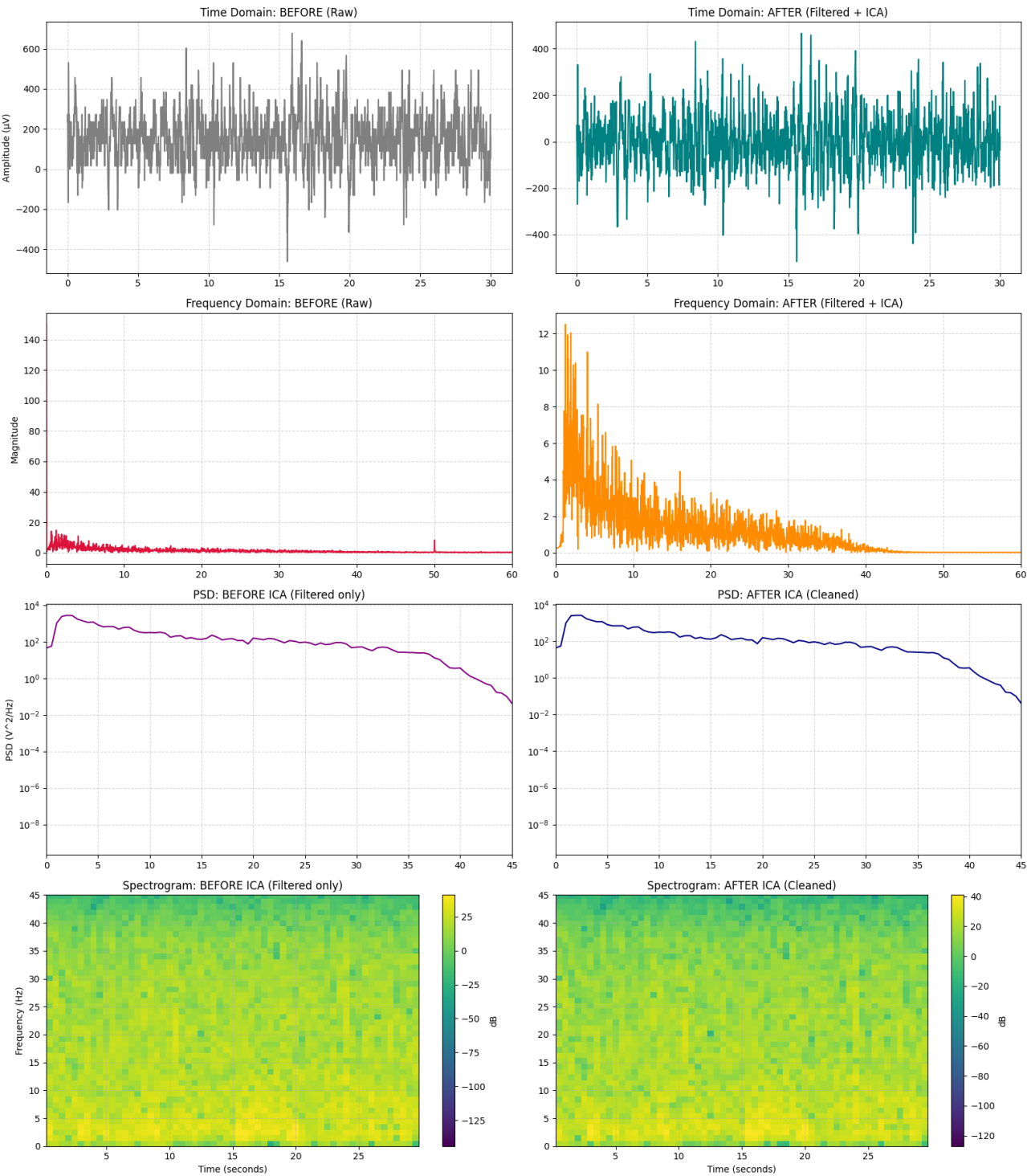
Channel: P3 | Group: ADHD_Negative



Channel: O2

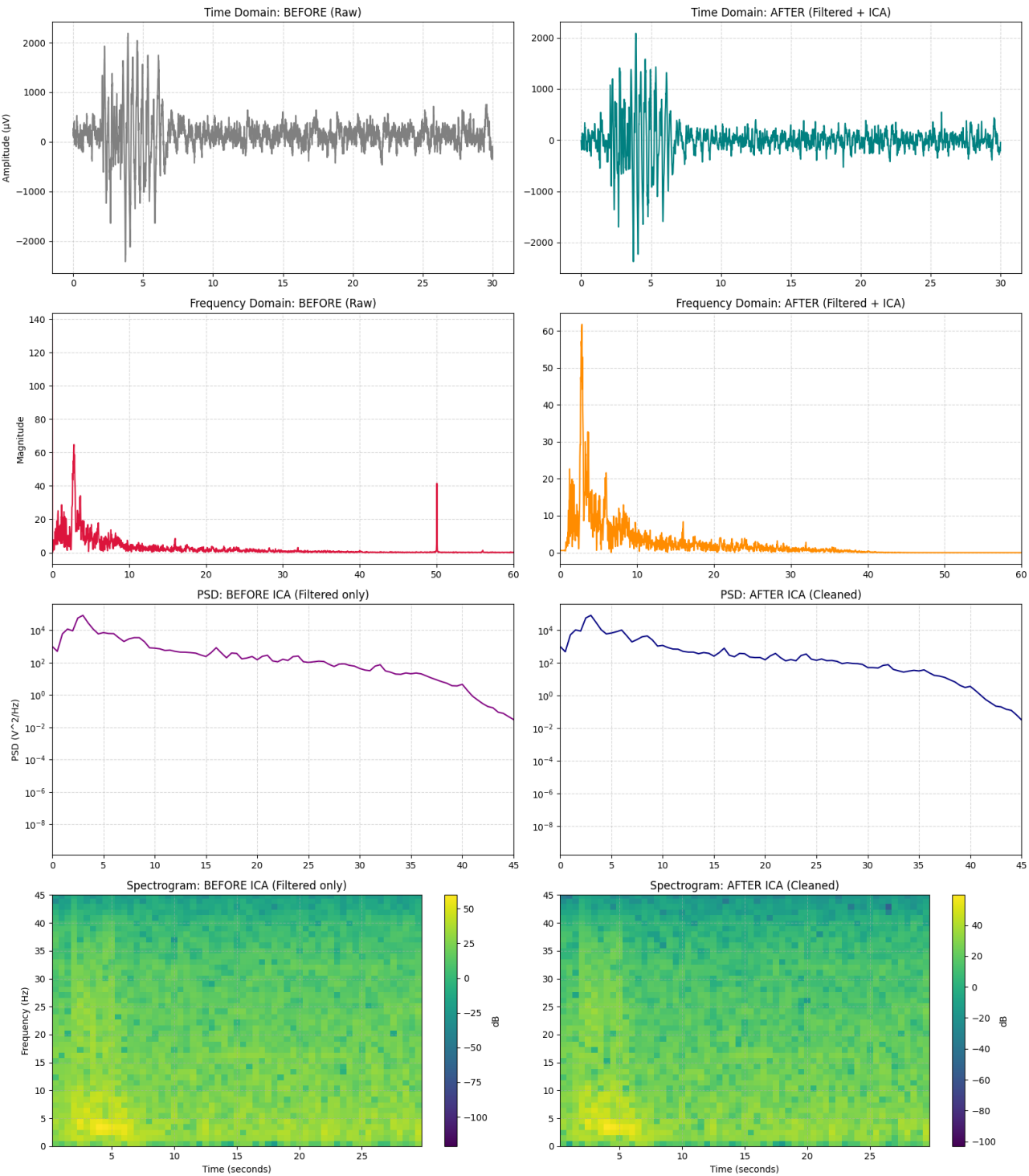
ADHD Positive — Channel O2

Channel: O2 | Group: ADHD_Positive



ADHD Negative (Control) — Channel O2

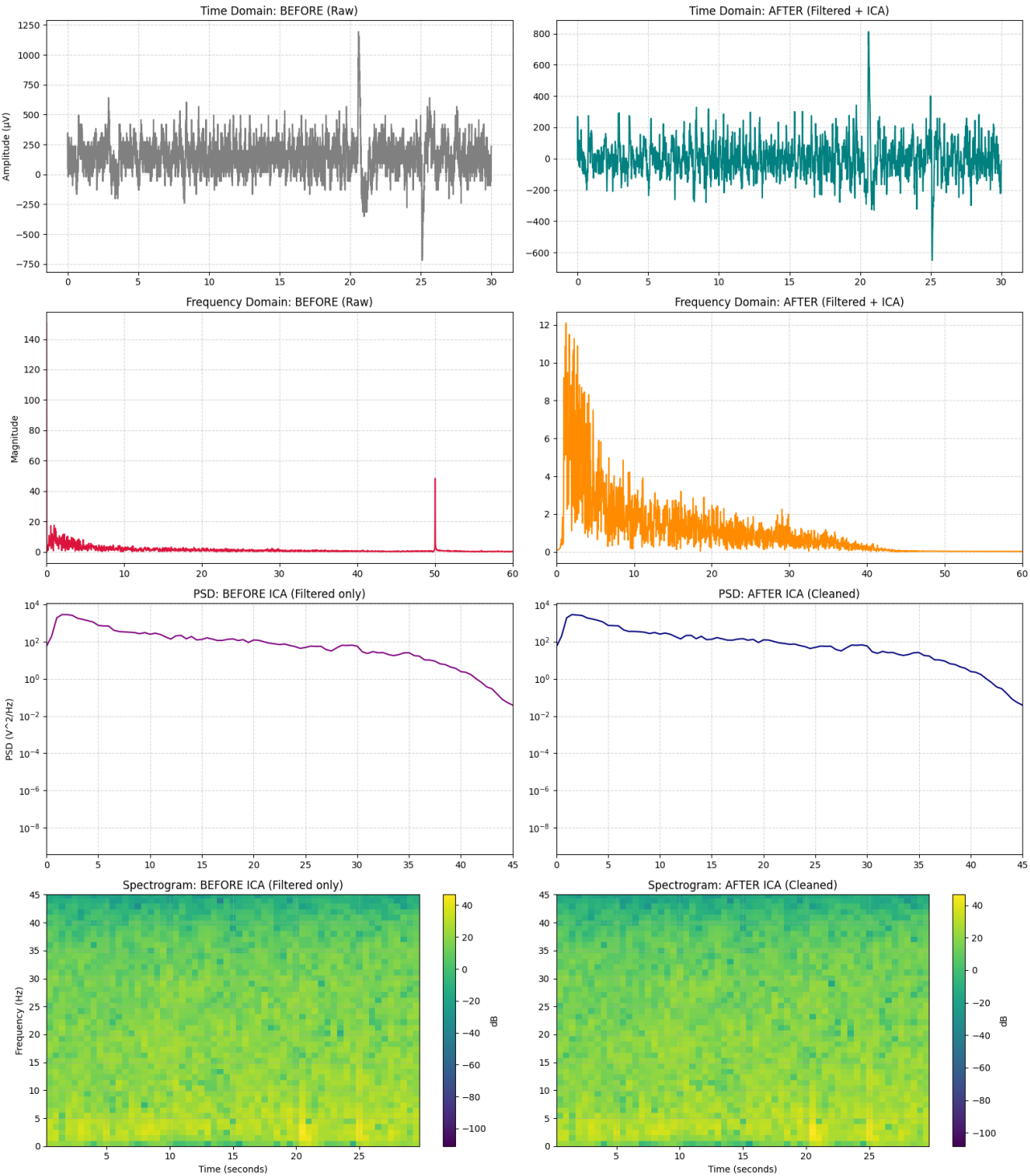
Channel: O2 | Group: ADHD_Negative



Channel: O1

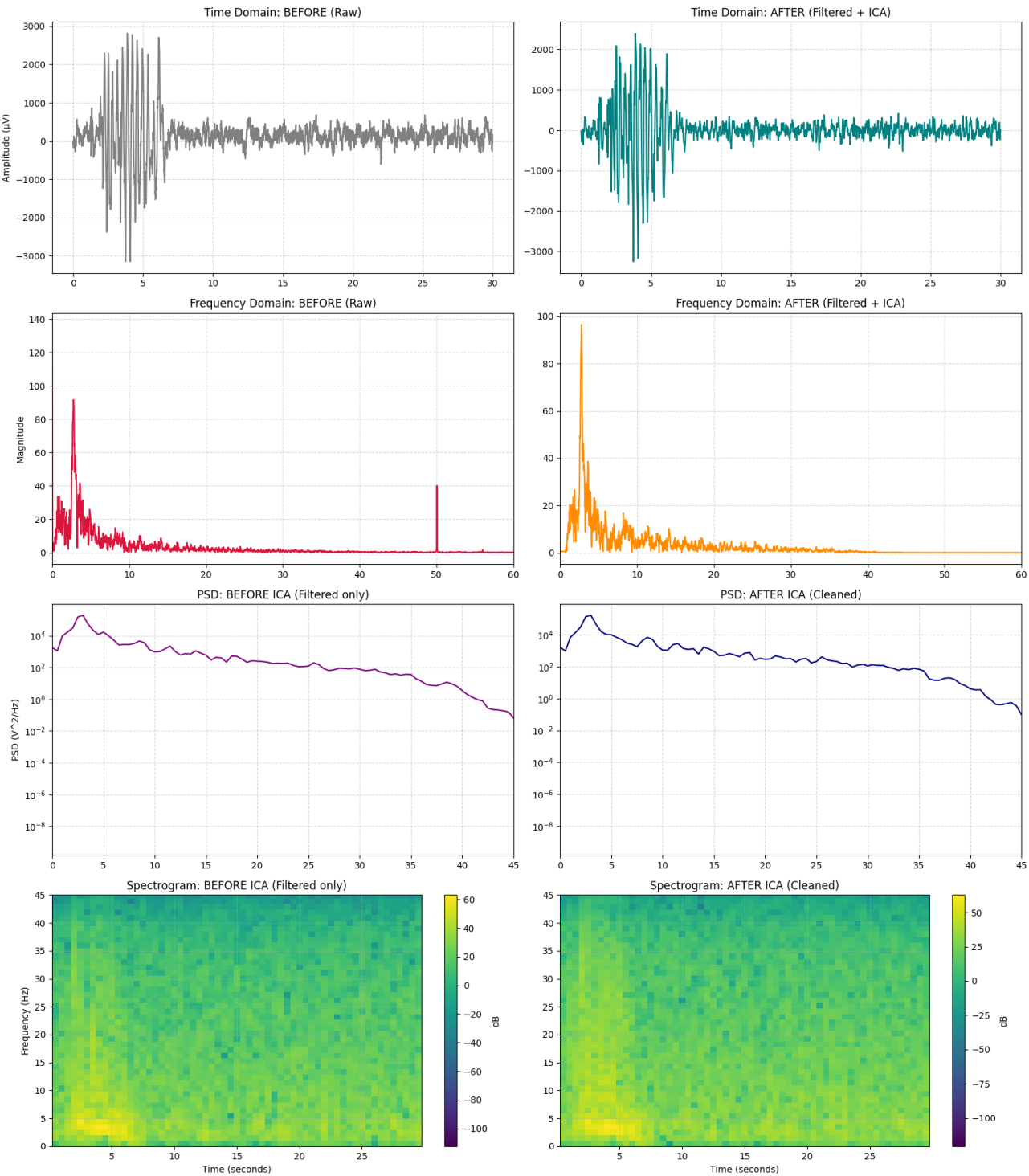
ADHD Positive — Channel O1

Channel: O1 | Group: ADHD_Positive



ADHD Negative (Control) — Channel O1

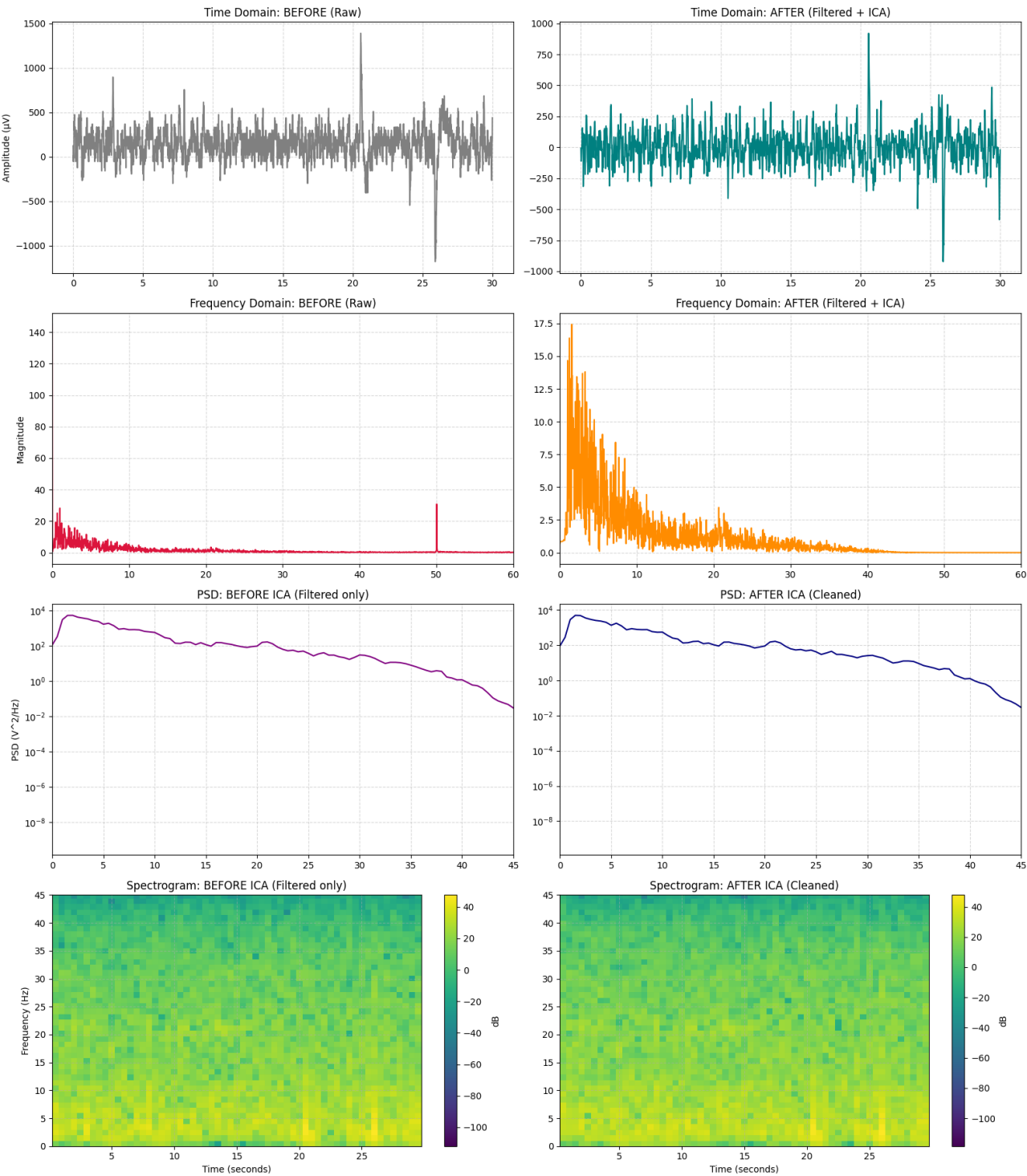
Channel: O1 | Group: ADHD_Negative



Channel: Fz

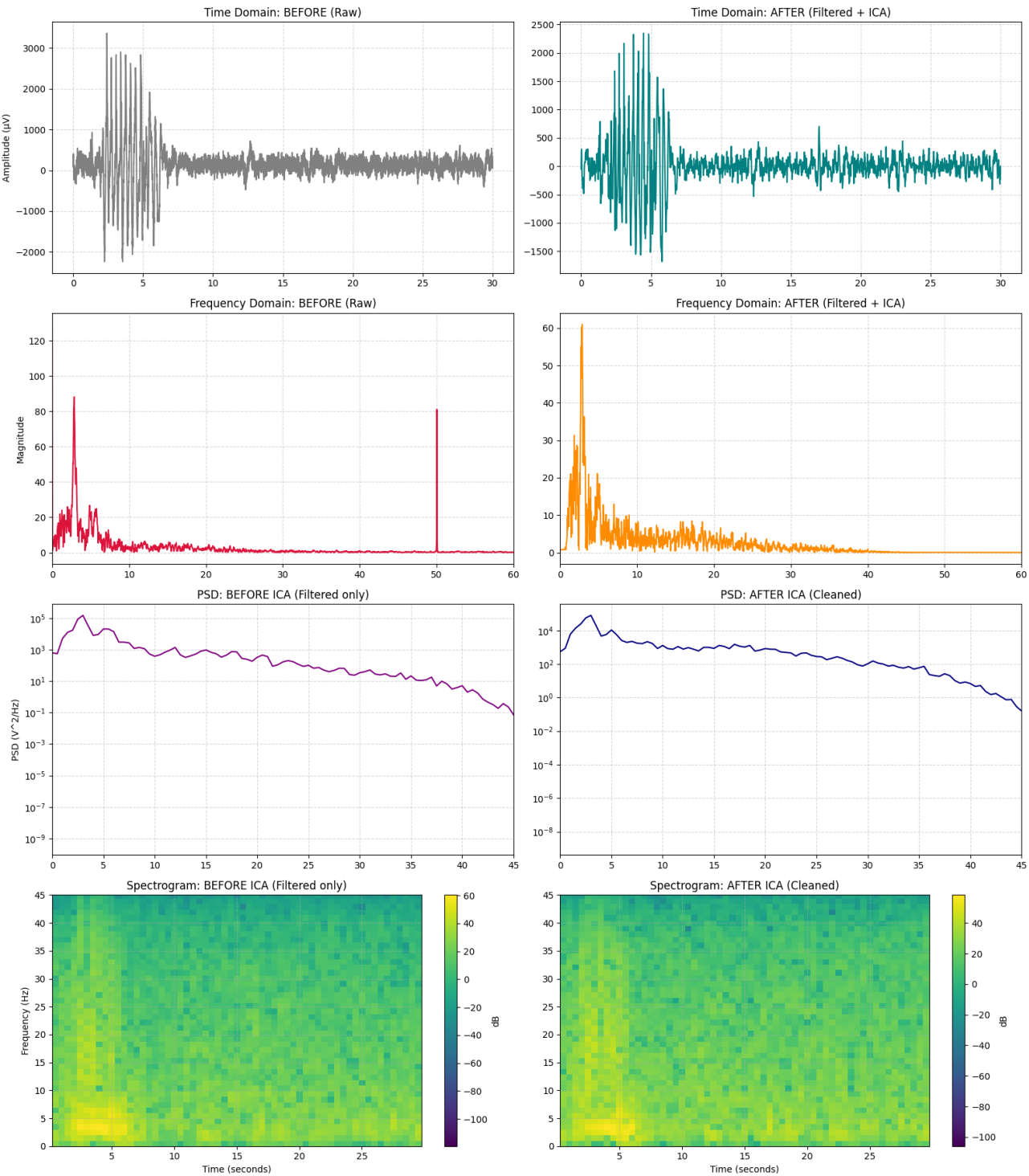
ADHD Positive — Channel Fz

Channel: Fz | Group: ADHD_Positive



ADHD Negative (Control) — Channel Fz

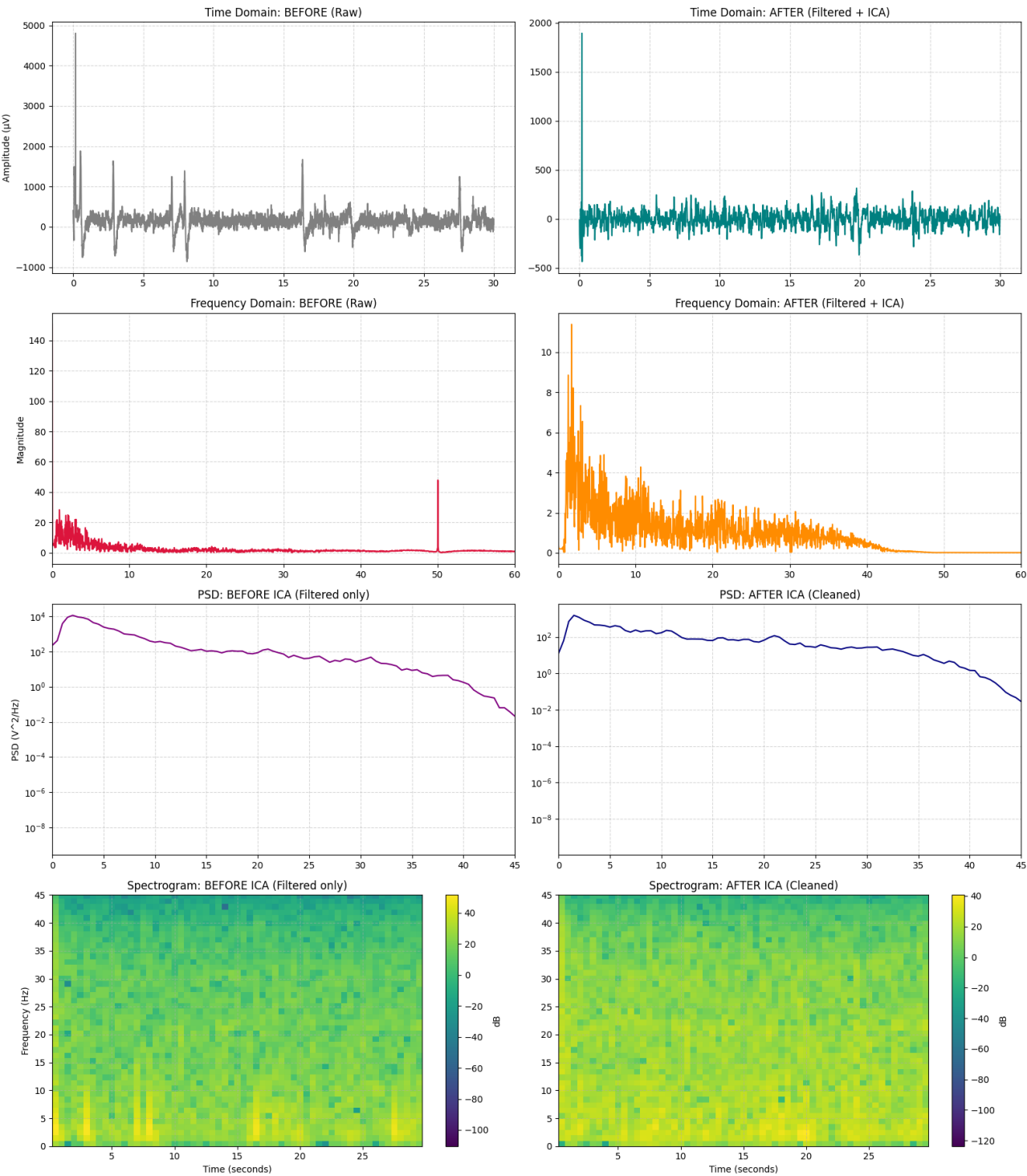
Channel: Fz | Group: ADHD_Negative



Channel: Fp2

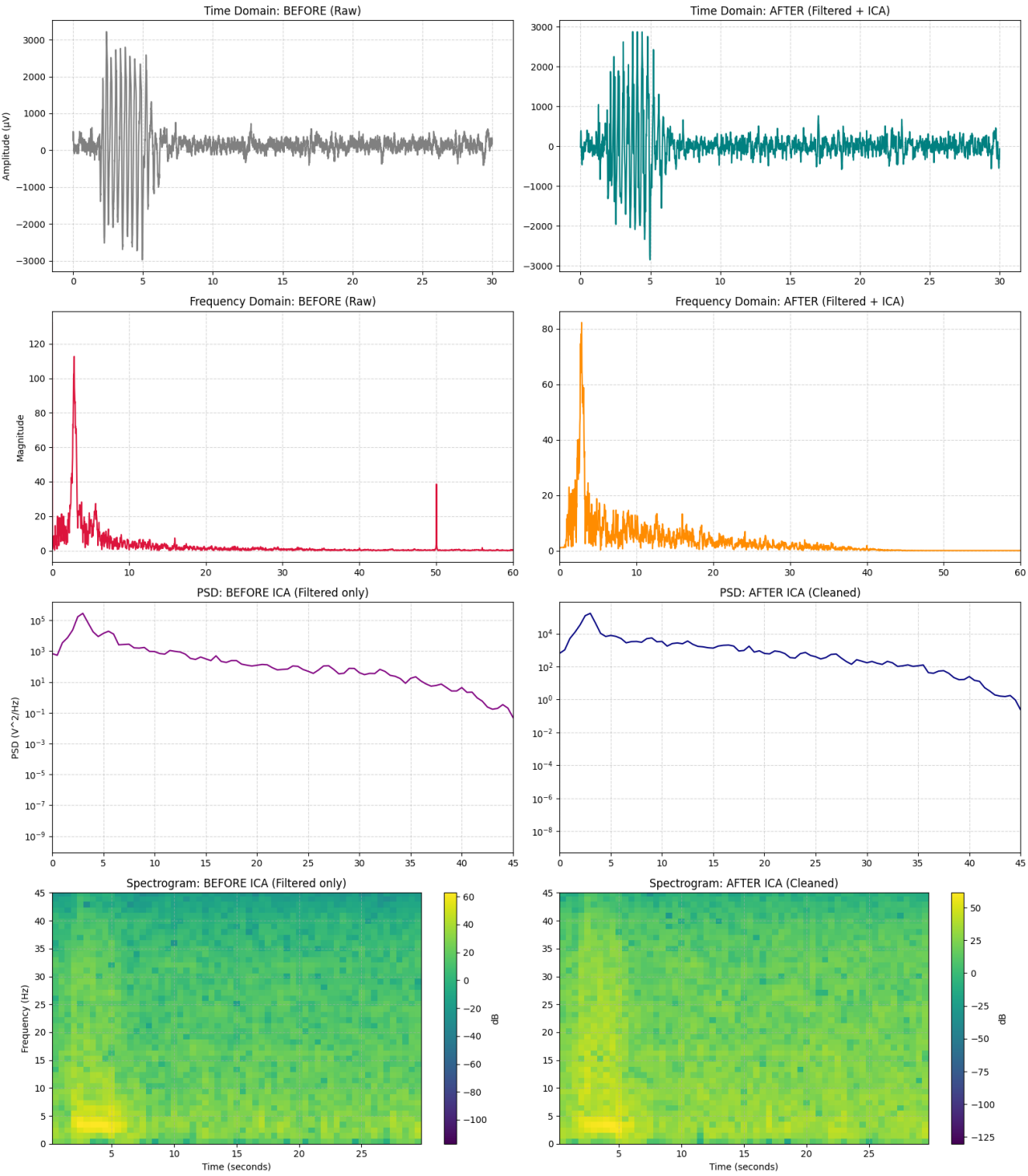
ADHD Positive — Channel Fp2

Channel: Fp2 | Group: ADHD_Positive



ADHD Negative (Control) — Channel Fp2

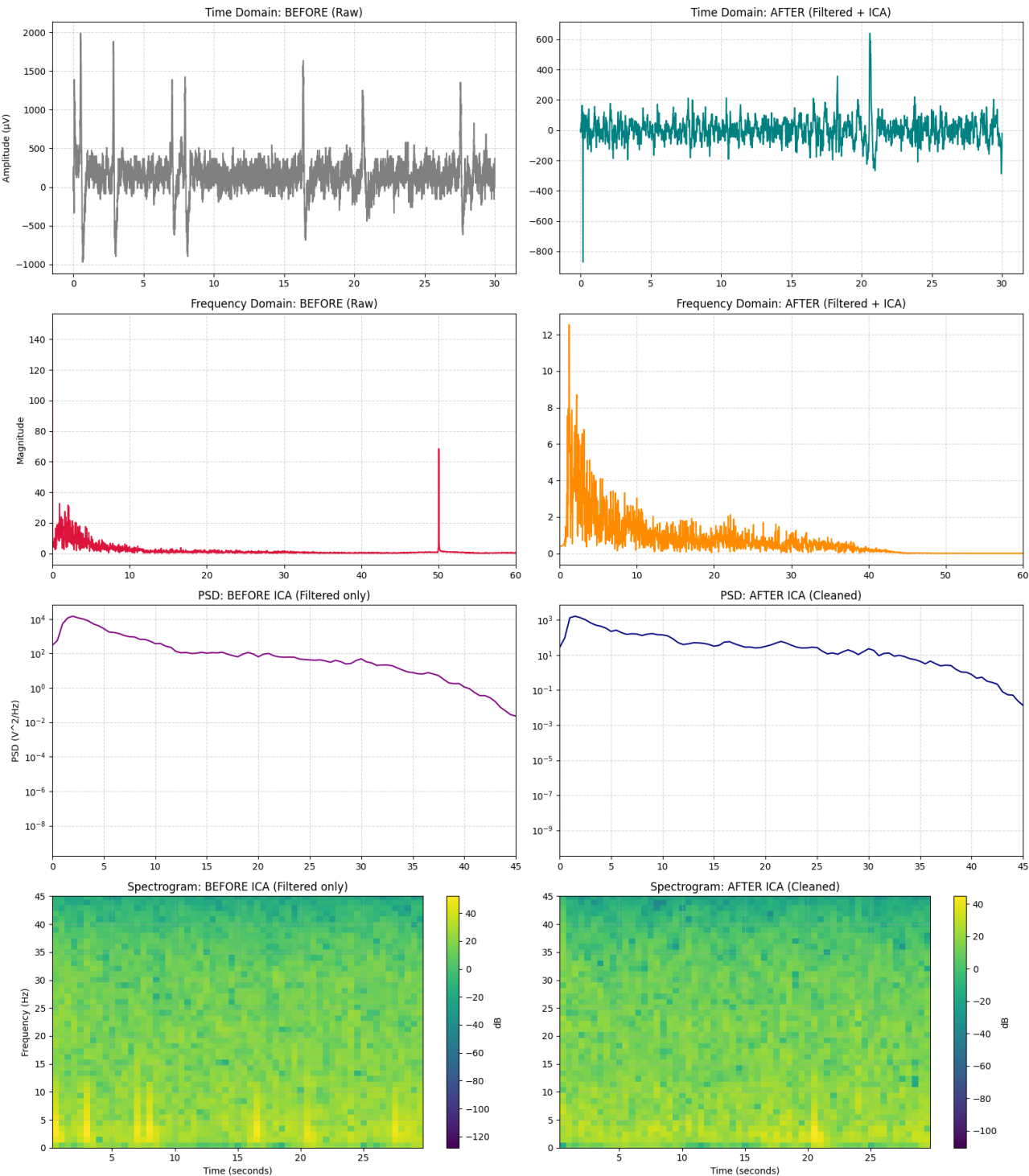
Channel: Fp2 | Group: ADHD_Negative



Channel: Fp1

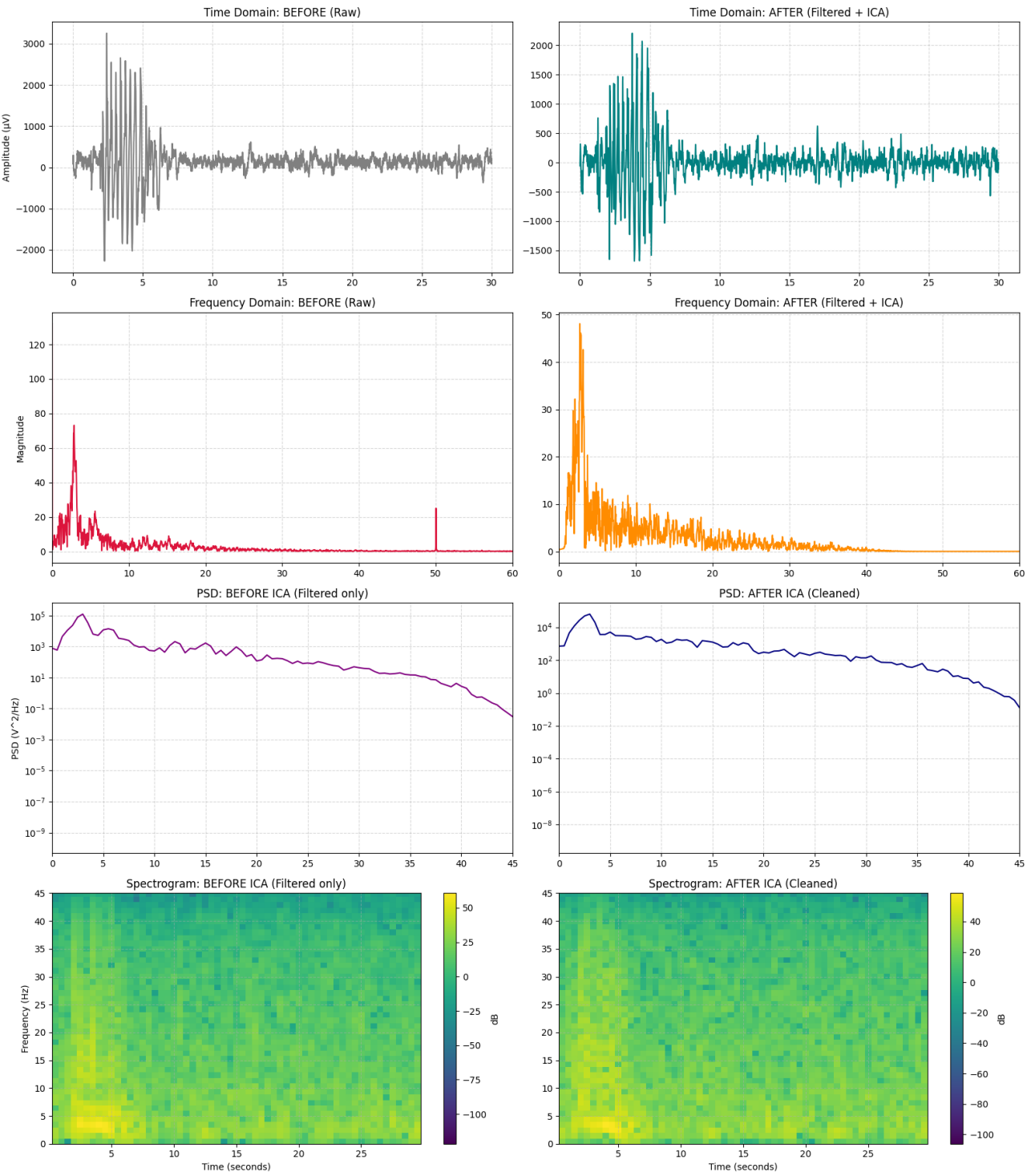
ADHD Positive — Channel Fp1

Channel: Fp1 | Group: ADHD_Positive



ADHD Negative (Control) — Channel Fp1

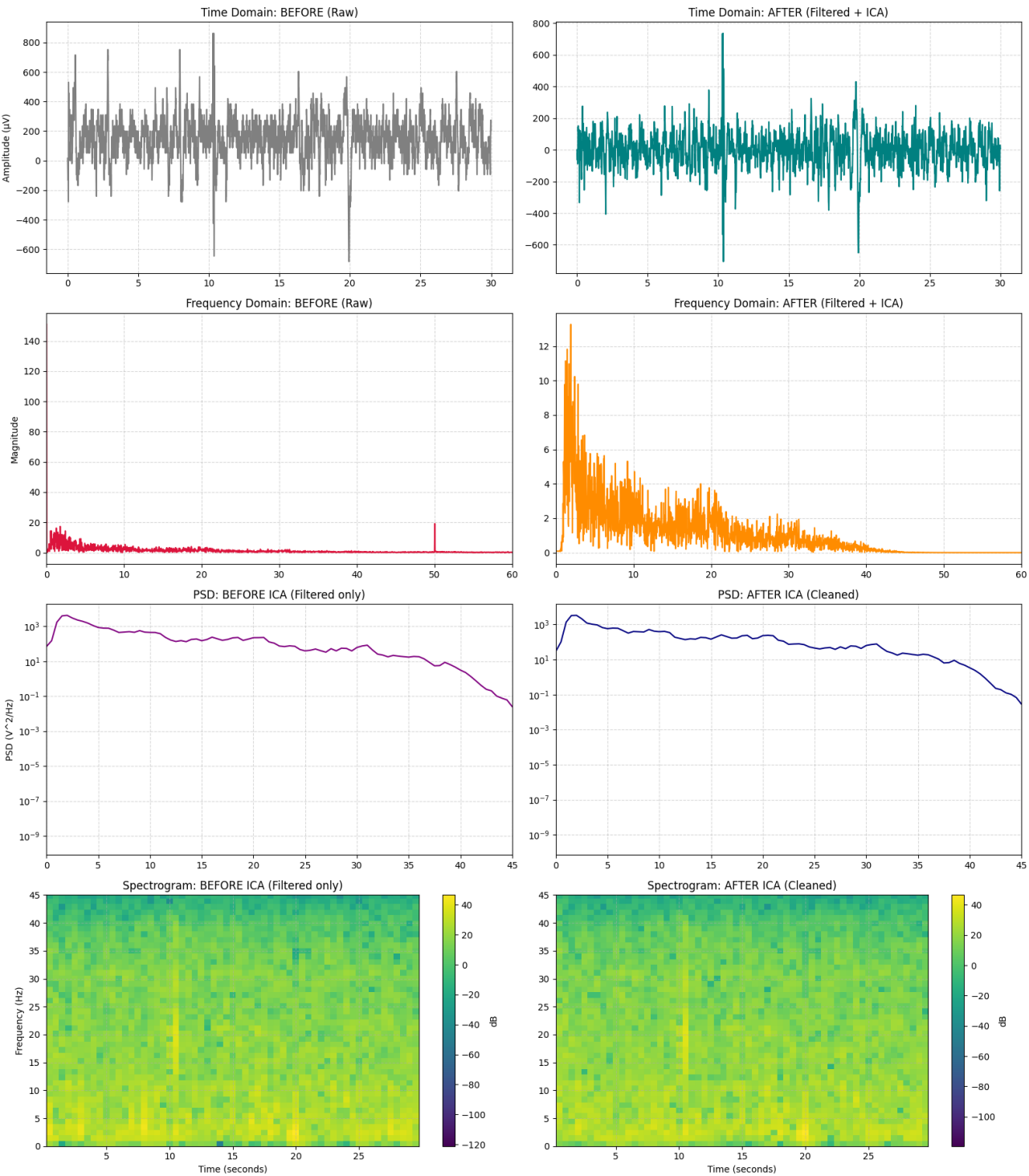
Channel: Fp1 | Group: ADHD_Negative



Channel: F8

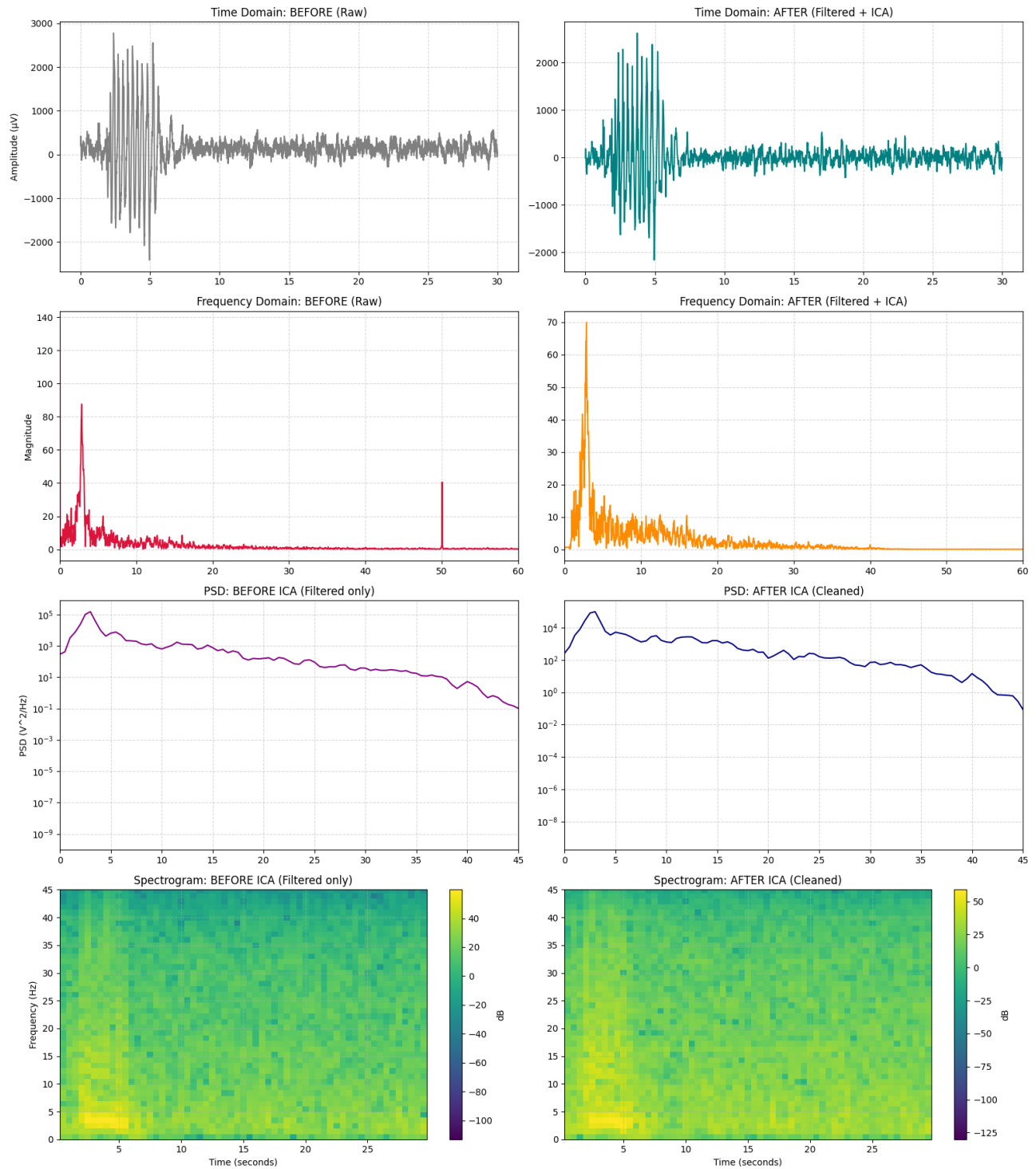
ADHD Positive — Channel F8

Channel: F8 | Group: ADHD_Positive



ADHD Negative (Control) — Channel F8

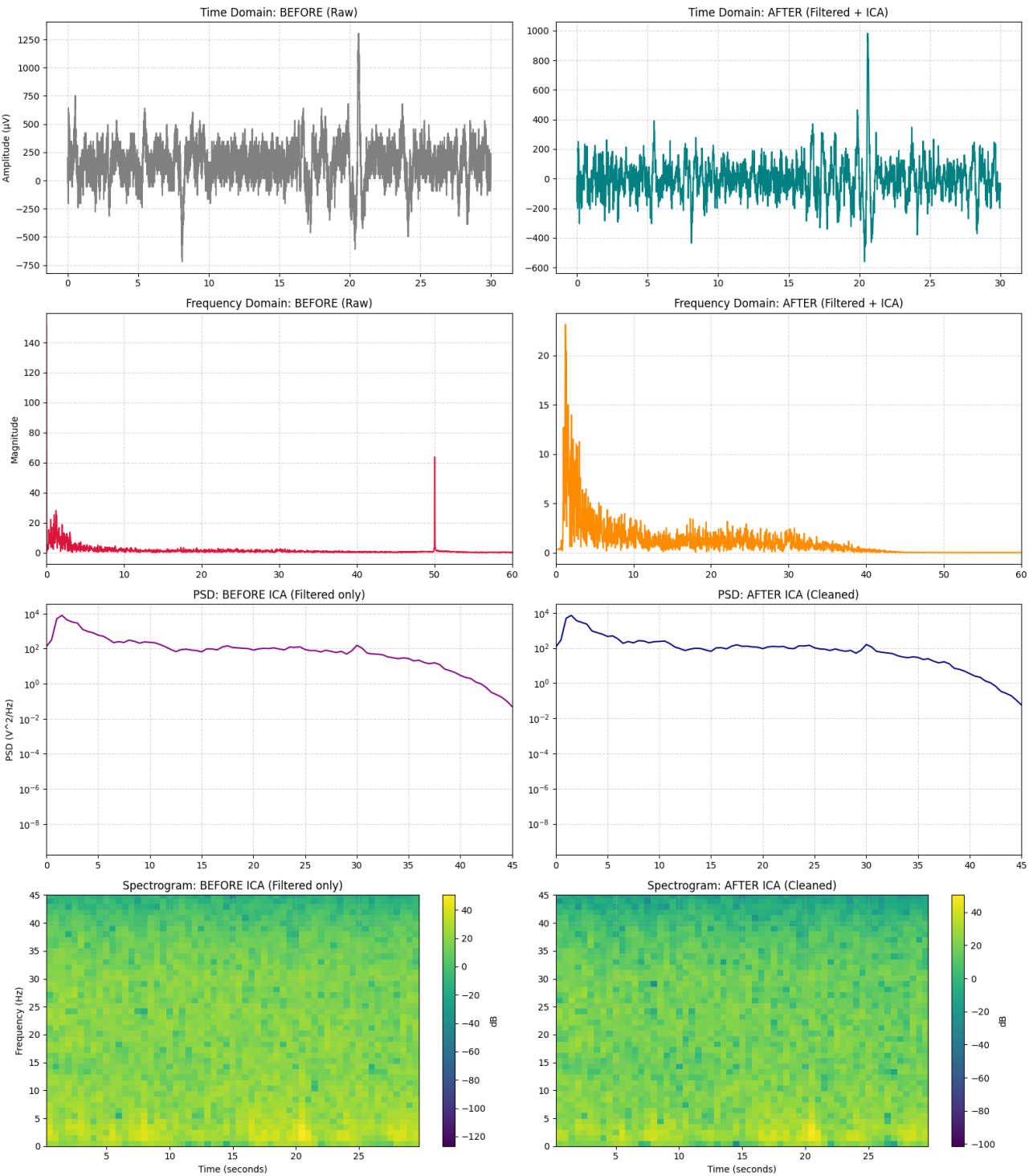
Channel: F8 | Group: ADHD_Negative



Channel: F7

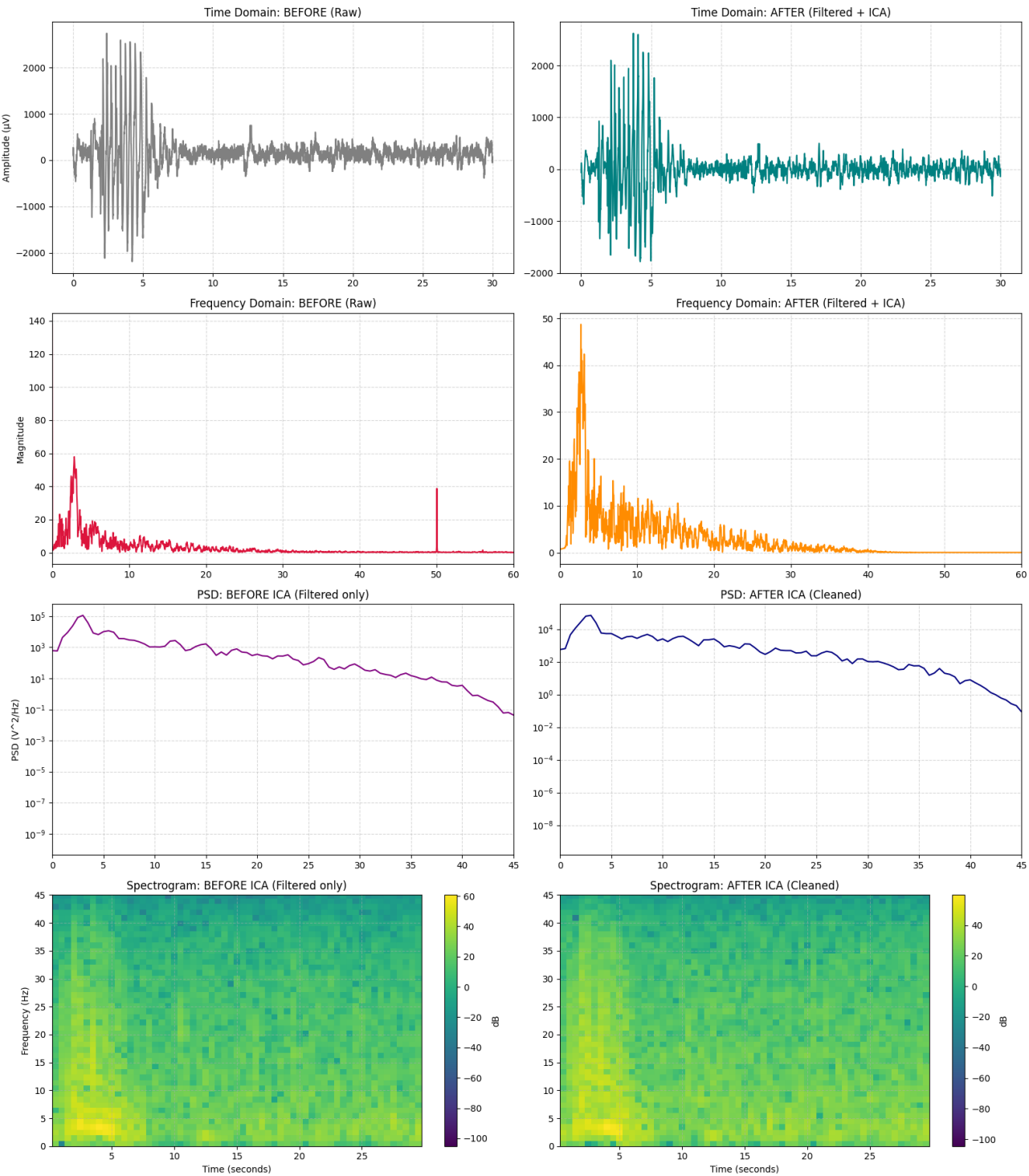
ADHD Positive — Channel F7

Channel: F7 | Group: ADHD_Positive



ADHD Negative (Control) — Channel F7

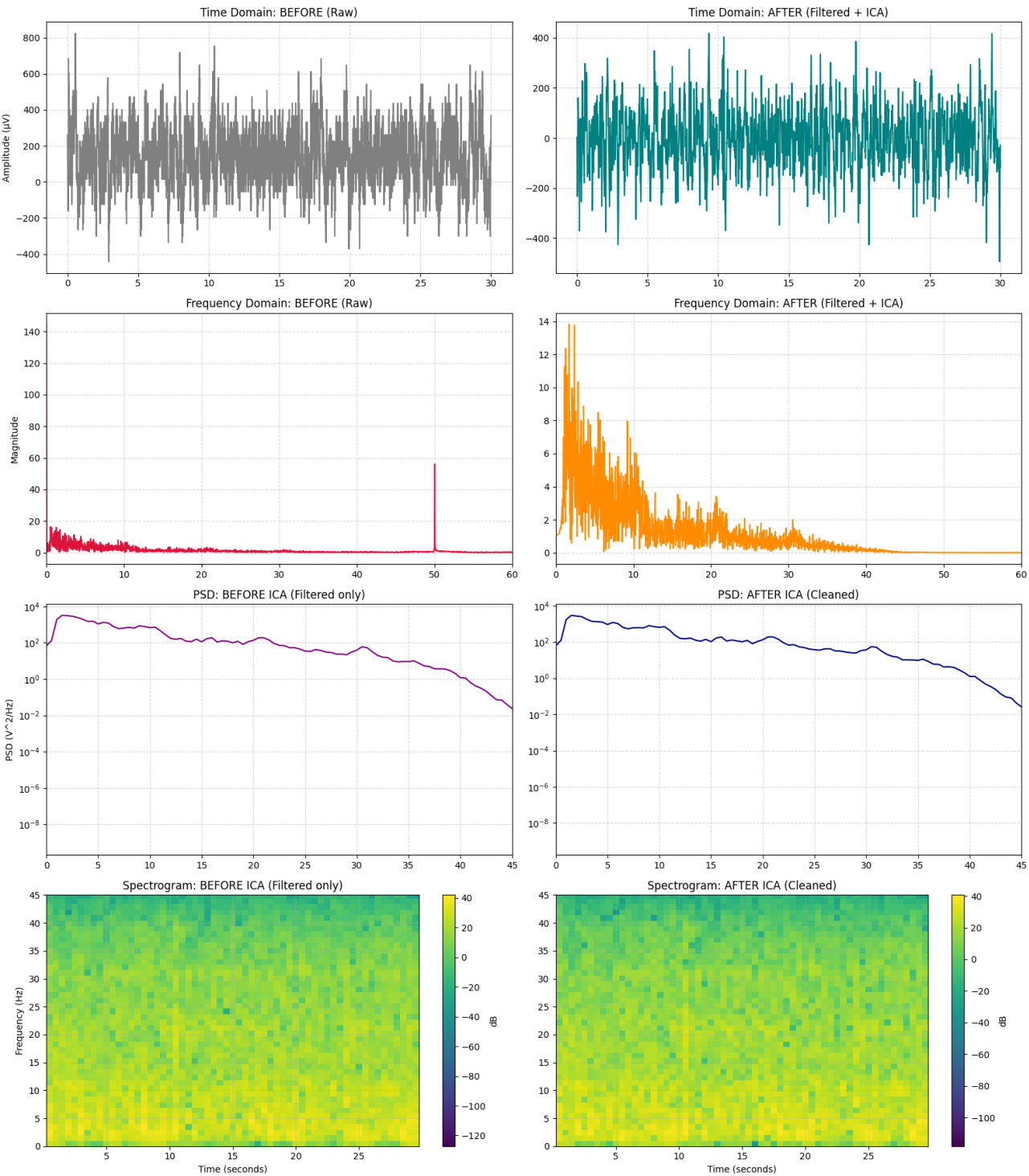
Channel: F7 | Group: ADHD_Negative



Channel: F4

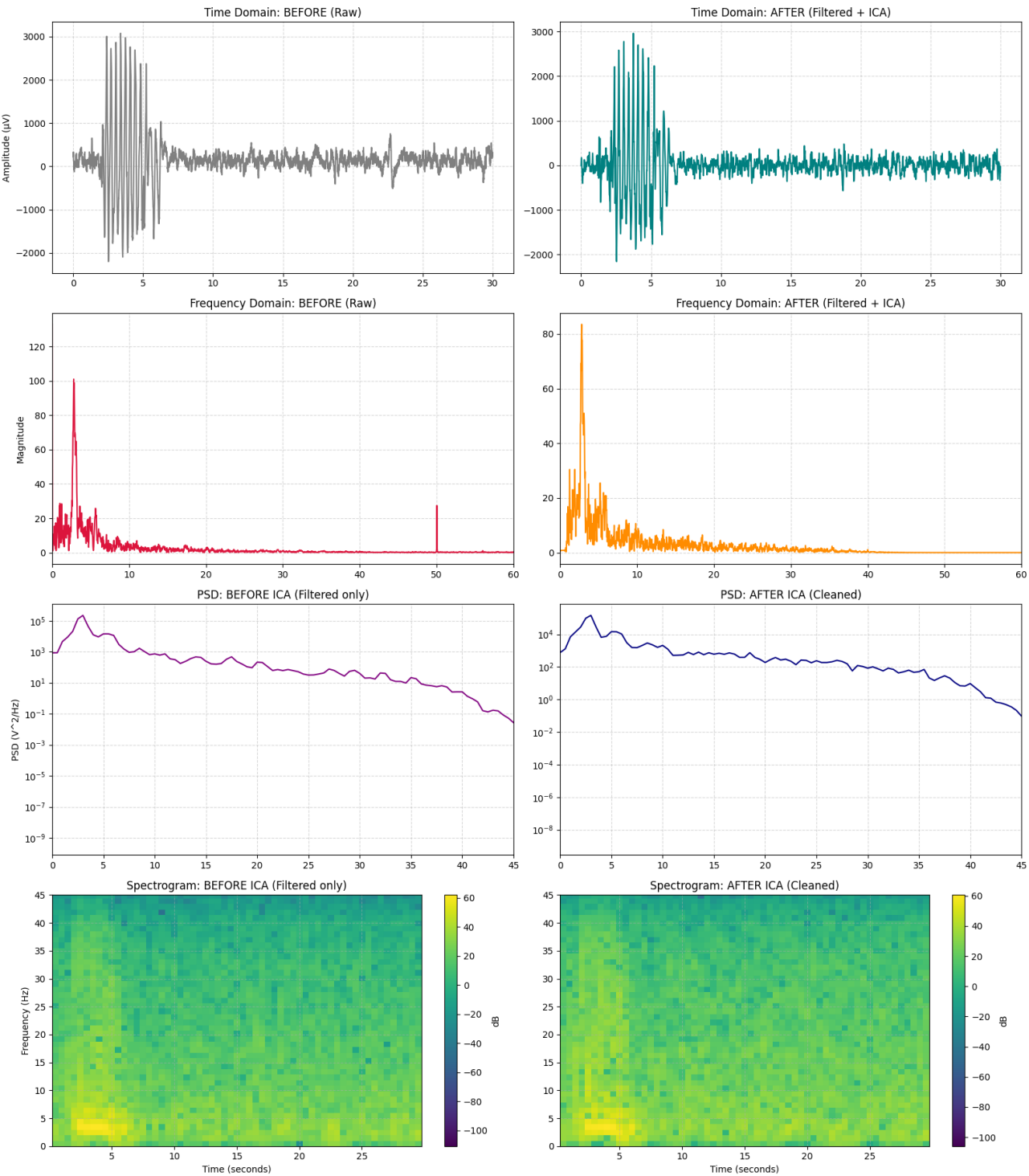
ADHD Positive — Channel F4

Channel: F4 | Group: ADHD_Positive



ADHD Negative (Control) — Channel F4

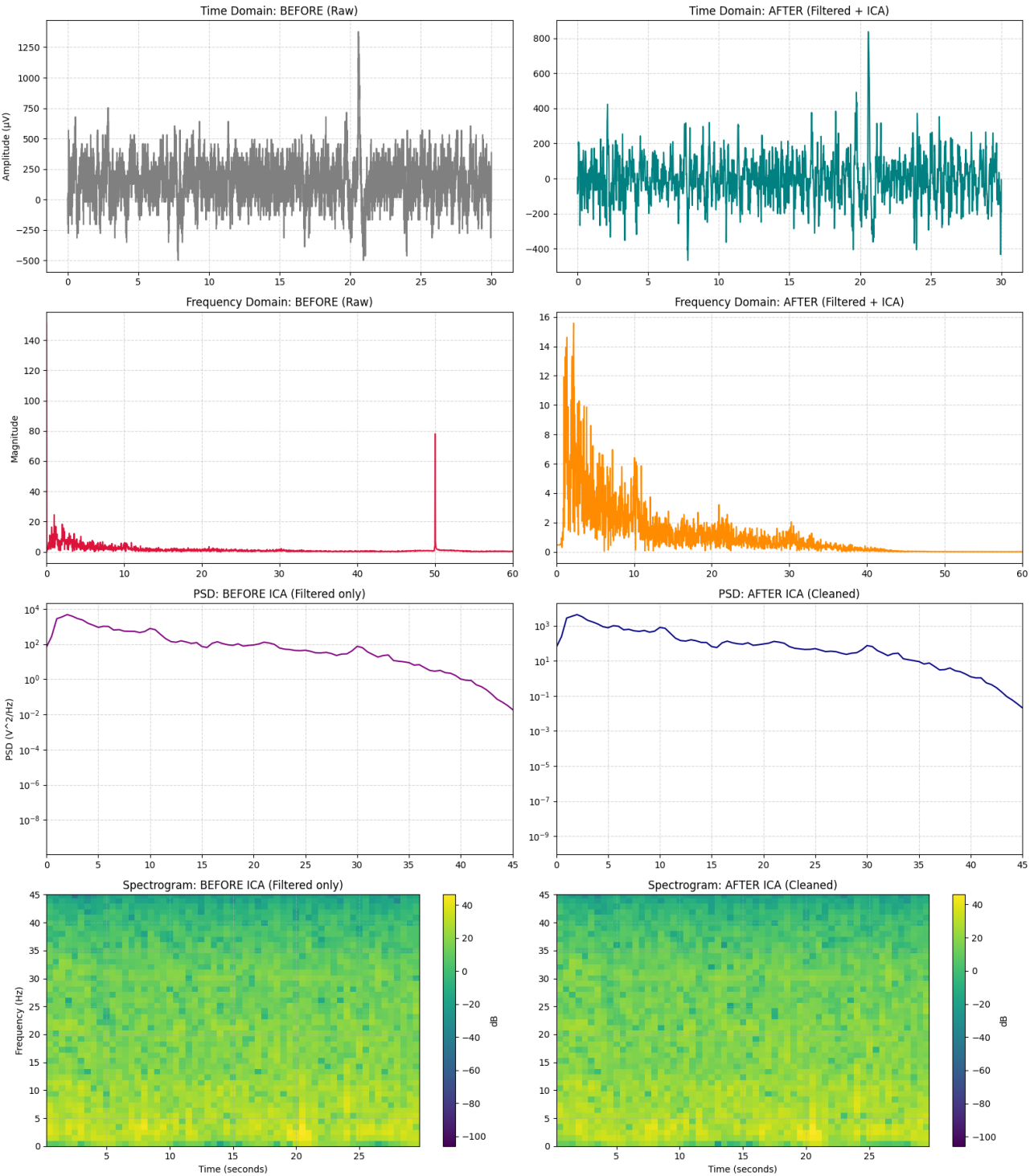
Channel: F4 | Group: ADHD_Negative



Channel: F3

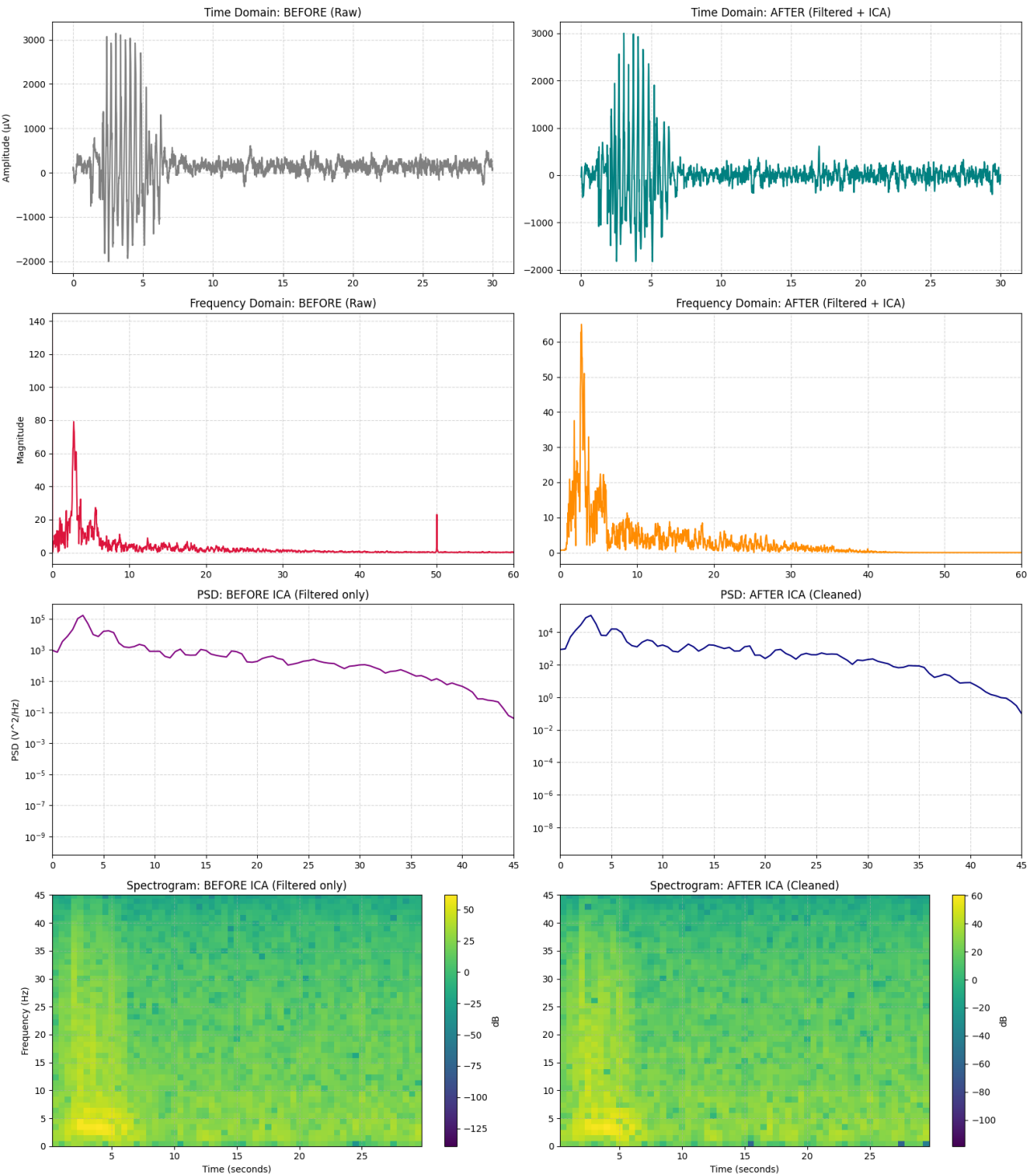
ADHD Positive — Channel F3

Channel: F3 | Group: ADHD_Positive



ADHD Negative (Control) — Channel F3

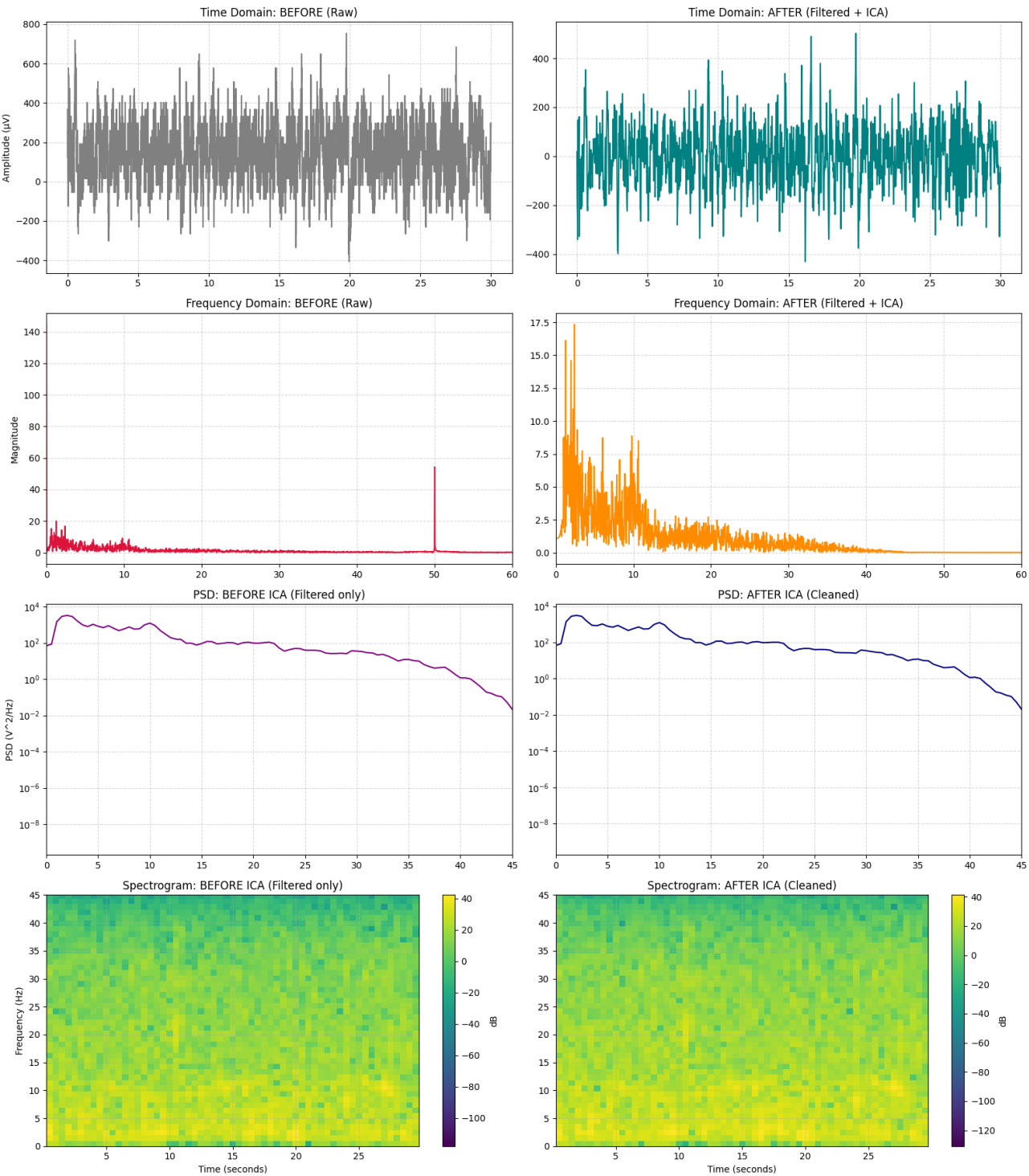
Channel: F3 | Group: ADHD_Negative



Channel: Cz

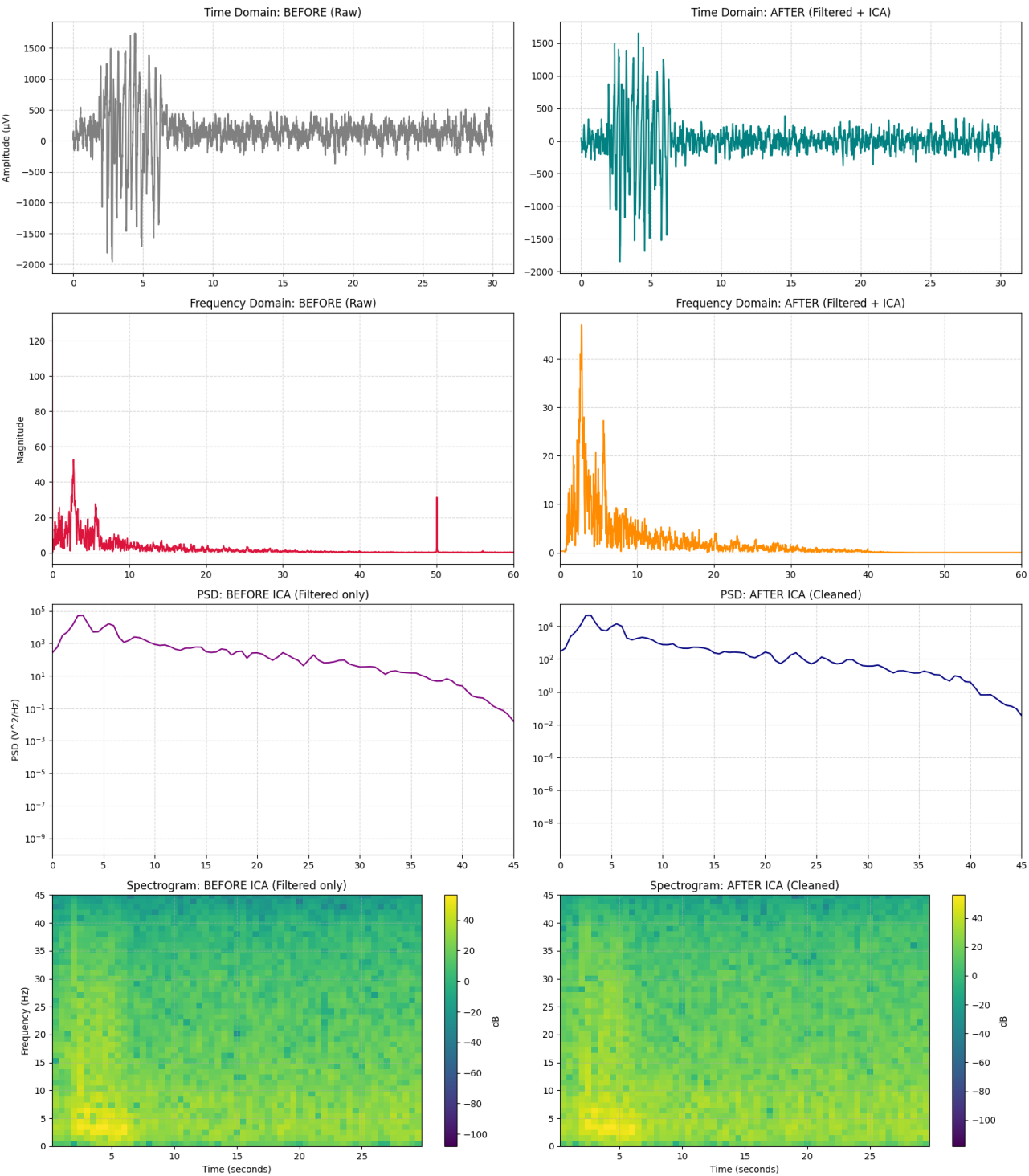
ADHD Positive — Channel Cz

Channel: Cz | Group: ADHD_Positive



ADHD Negative (Control) — Channel Cz

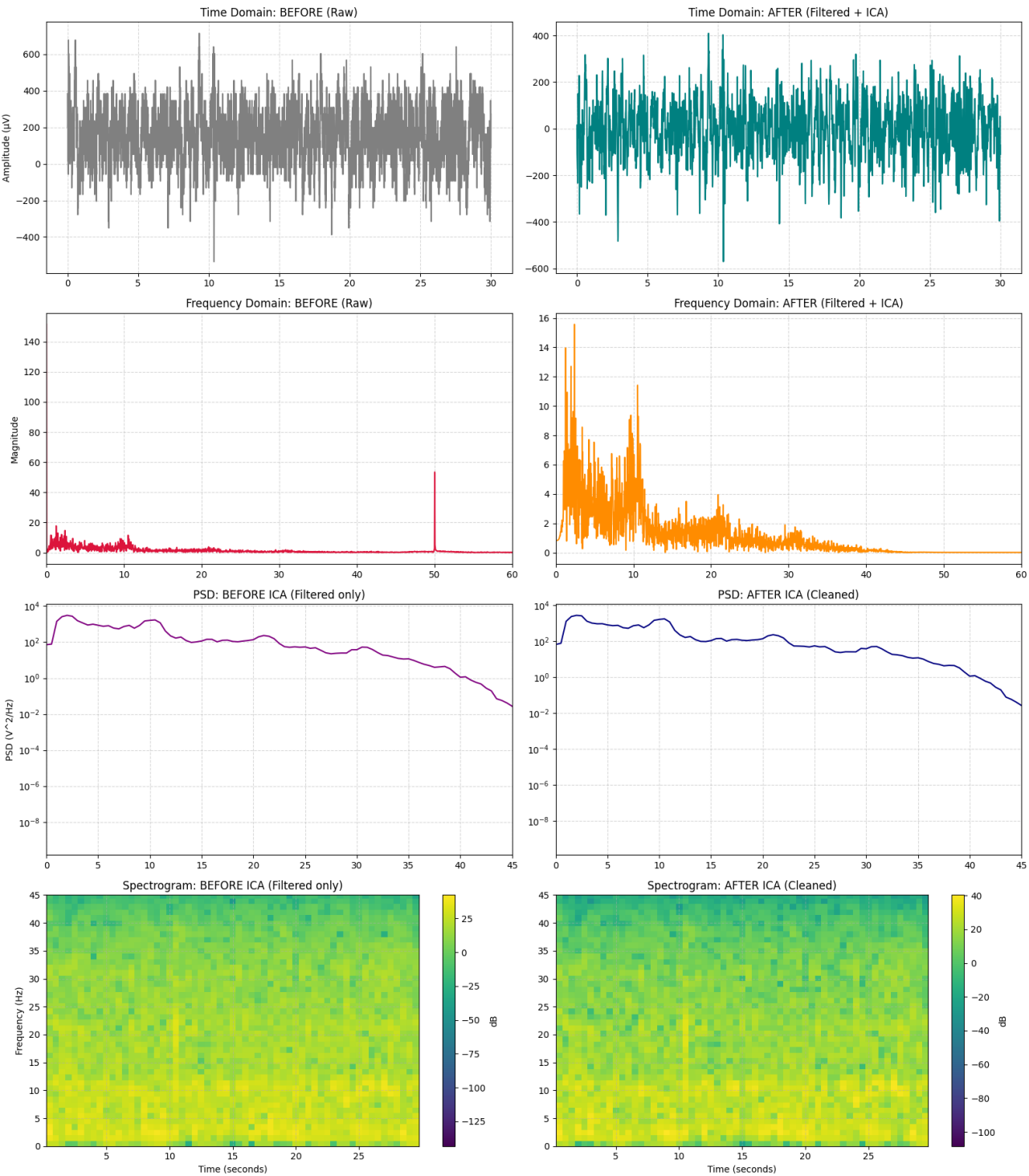
Channel: Cz | Group: ADHD_Negative



Channel: C4

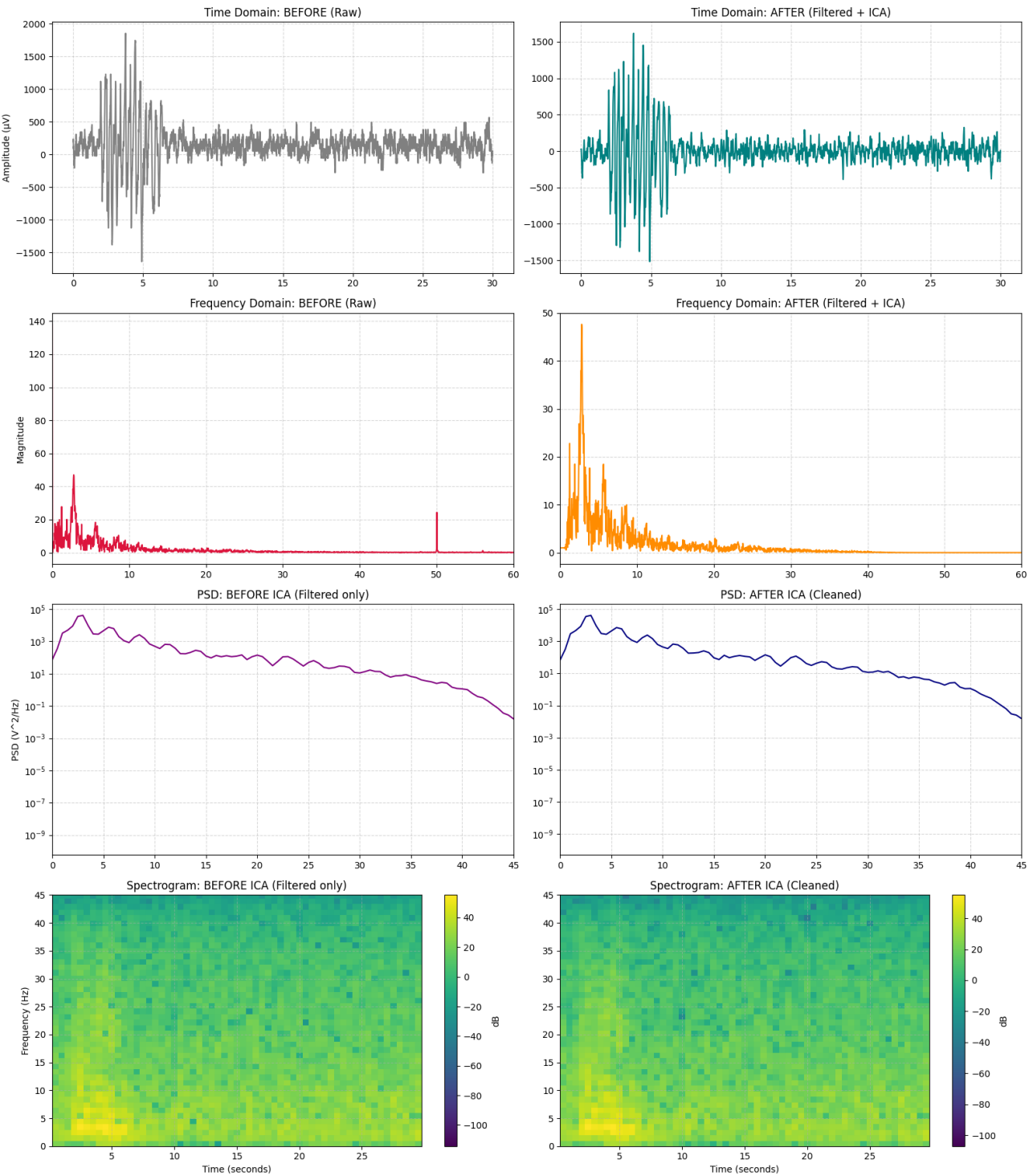
ADHD Positive — Channel C4

Channel: C4 | Group: ADHD_Positive



ADHD Negative (Control) — Channel C4

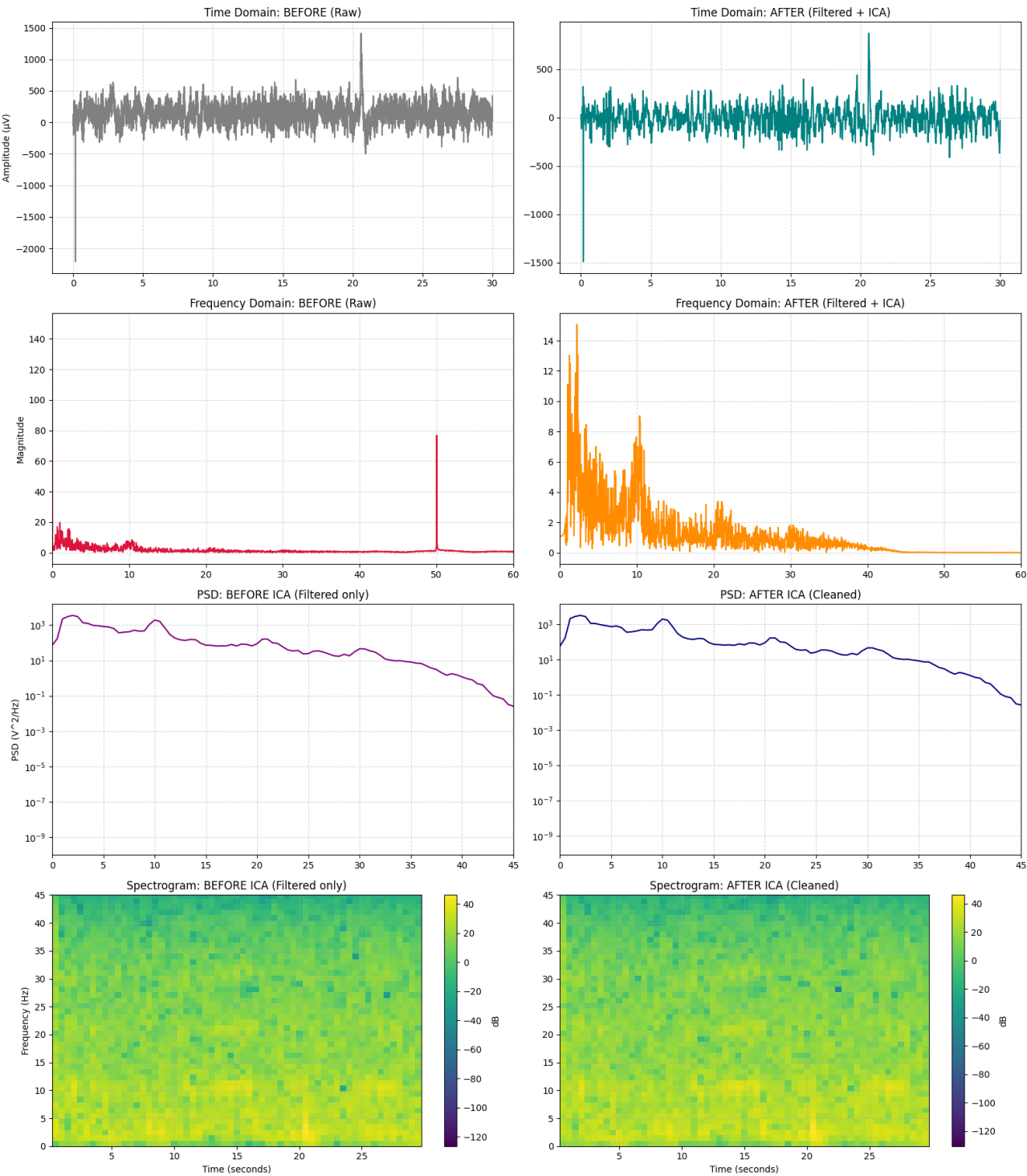
Channel: C4 | Group: ADHD_Negative



Channel: C3

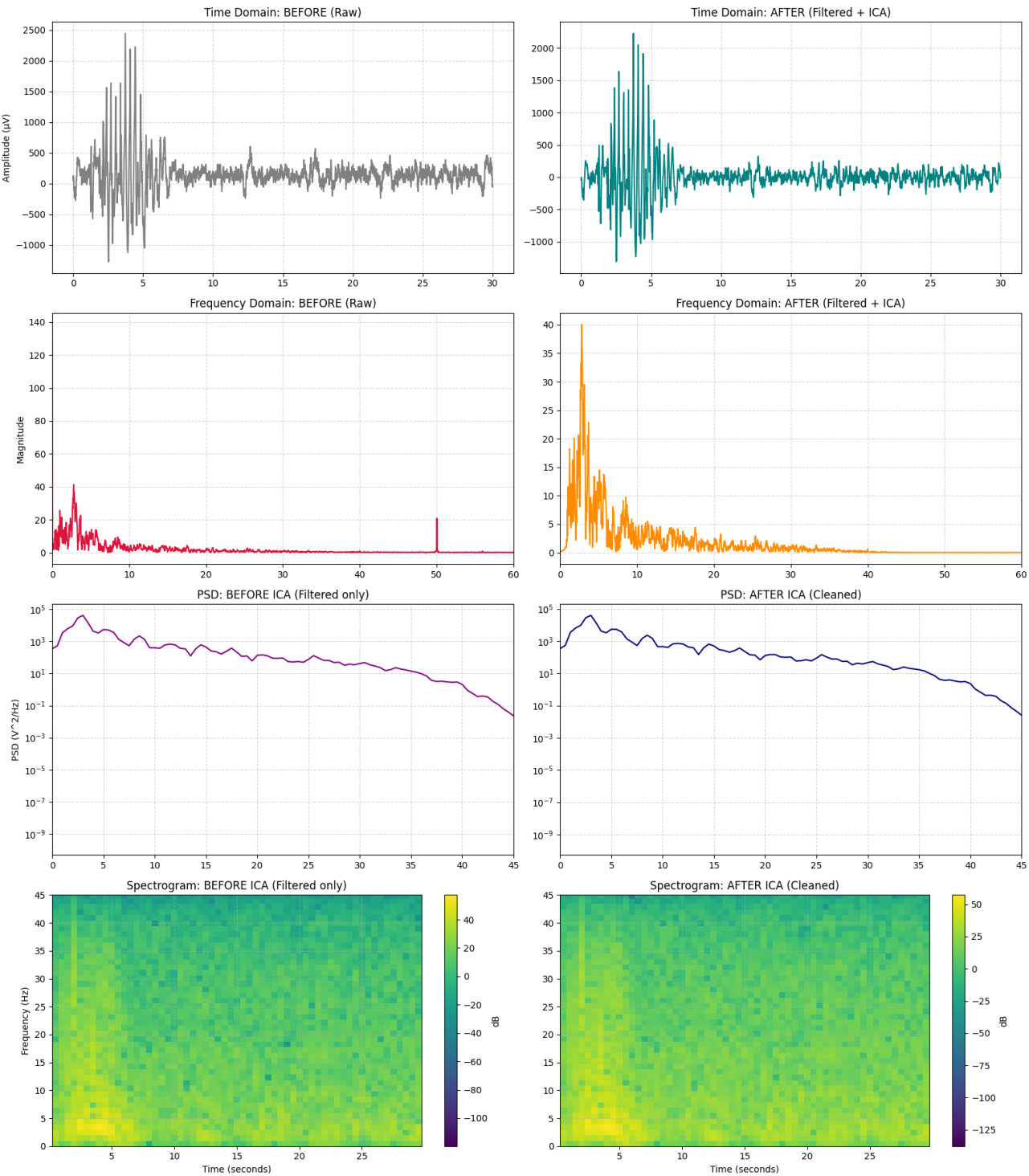
ADHD Positive — Channel C3

Channel: C3 | Group: ADHD_Positive



ADHD Negative (Control) — Channel C3

Channel: C3 | Group: ADHD_Negative



Analysis Observation:

1. Signals Detected in the ADHD_Negative Group

The 30-second epoch for this group is dominated by a single, massive physical disturbance rather than standard resting baseline activity.

- **Massive Global Event:** There is a continuous, extreme amplitude burst occurring between the 2.0 and 6.0-second marks.
- **Extreme Voltage:** The peak-to-peak amplitudes during this 4-second window reach up to $\pm 3000 \mu\text{V}$, which is exponentially higher than typical neural activity.
- **Widespread Distribution:** This massive signal deflection is not localized; it is registered across all 19 channels, from the extreme frontal (Fp1) to the extreme occipital (O2) regions.
- **Frequency Impact:** The Spectrogram and Power Spectral Density (PSD) plots reveal that this 4-second event forces the vast majority of the signal's power into the lowest frequency range (0–5 Hz, Delta band).
- **Isolated P8 Anomaly:** On channel P8, there is a distinct, sharp downward spike reaching approximately $-2000 \mu\text{V}$ at the 16-second mark, which is not present in the surrounding channels.

2. Signals Detected in the ADHD_Positive Group

The raw data in this epoch shows a much quieter, more standard baseline, but it is characterized by frequent, rapid, and repeating signal spikes.

- **Stable Baseline:** The underlying baseline activity generally remains within a much tighter $\pm 500 \mu\text{V}$ range.
- **Rapid Frontal Transients:** The signal is regularly interrupted by sharp, brief spikes occurring roughly at the 0.5s, 5s, 11s, 16s, and 26s marks.
- **Frontal Dominance:** These rapid spikes are highly pronounced in the frontal channels (Fp1, Fp2, F7, F8) and gradually fade in amplitude in the channels closer to the back of the head.
- **The 20.5-Second Anomaly:** There is a highly distinct, sharp negative deflection occurring at exactly 20.5 seconds. This specific spike is prominent across multiple channels (C3, F3, F7, and P7) and leaves a severe negative drop (exceeding $-1500 \mu\text{V}$ on C3) even after processing.

3. Key Differences Between the Groups

- **Duration and Morphology:** The Negative group is defined by one sustained, rolling 4-second wave of extreme voltage. In contrast, the Positive group is defined by multiple, split-second, sharp transients interrupting a tighter baseline.
- **Spatial Spread:** The massive 4-second event in the Negative group is global, appearing with high intensity across the entire scalp. The recurring rapid spikes in the Positive group are heavily localized to the front of the head.
- **Impact on the Frequency Spectrum:** The Negative group's long event artificially crushes the power spectrum into the lowest frequencies (Delta). The Positive group's rapid spikes create brief, vertical broadband bursts of energy across the spectrograms rather than a single sustained low-frequency block.

4. Processing Pipeline Detections

Looking at the "BEFORE" and "AFTER" plots reveals how the processing steps impacted the two distinct groups:

- **50 Hz Signal Removal:** In the raw frequency plots for both groups, a sharp, artificial peak is detected at exactly 50 Hz. The "AFTER" plots show this 50 Hz spike was entirely successfully eliminated across all 38 channels.
- **Algorithmic Impact:** The Independent Component Analysis (ICA) algorithm reacted differently to the two groups. In the Positive group, the algorithm successfully reduced the amplitude of the rapid frontal spikes. However, in the Negative group, the algorithm failed to resolve the massive 4-second global event, leaving the "cleaned" output highly distorted.

code link:

https://github.com/rith1509/Bio-Signal-and-Image-Analysis/blob/main/adhd_mywork.ipynb